# Partial Differential Equations

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# Introduction

Partial differential equations arise naturally when a physical quantity depends on several independent variables, most commonly space and time. Unlike an ordinary differential equation, which describes the evolution of a finite-dimensional state, a partial differential equation describes the evolution or equilibrium of an entire field.
Typical examples include the density of a moving material, the temperature of a body, the displacement of a vibrating string, the pressure of a fluid, and an electric or magnetic field. The common feature is that the state of the system is represented by a function such as $u = u(t, x)$, where $t$ denotes time and $x$ denotes position.
The purpose of this course is not merely to manipulate partial derivatives. A PDE is usually a local mathematical expression of a physical principle. Conservation of mass produces transport equations, Fourier's law of heat conduction produces the heat equation, Newton's law produces the wave equation, and equilibrium often produces the Laplace or Poisson equation.
This introduction derives these basic equations heuristically from their physical origins. The derivations are not intended to be mathematically complete. Instead, they explain why the equations appearing throughout these notes have the forms that they do.

## 0.1 From Physical Laws to Partial Differential Equations

The fundamental idea behind many PDEs is simple: examine a very small region of space and compare what enters, what leaves, and what is created inside.
Suppose $u(t, x)$ denotes the density of some quantity at time $t$ and position $x$. This quantity could represent mass, probability, population, energy, or concentration. If $a < b$, then

$$\int_a^b u(t, x)\, dx$$

is the total amount contained in the interval $[a, b]$.
Let $J(t, x)$ denote its flux. With the convention that positive flux moves to the right, the rate of change of the total amount in $[a, b]$ is

$$\frac{d}{dt}\int_a^b u(t, x)\, dx = J(t, a) - J(t, b) + \int_a^b f(t, x)\, dx,$$

where $f(t, x)$ represents an internal source or sink.
Assuming sufficient smoothness, differentiation under the integral sign and the fundamental theorem of calculus give

$$\int_a^b \left(\partial_t u + \partial_x J - f\right)\, dx = 0.$$

Since this holds for every interval $[a, b]$, we obtain the local conservation law

$$\partial_t u + \partial_x J = f.$$

This passage from an integral balance law to a differential equation is one of the central mechanisms by which PDEs arise. The integral law describes a finite region, while the PDE records the same physical law at every point.
In several spatial dimensions, if $J(t, x) \in \mathbb{R}^d$ is the flux vector, the same principle becomes

$$\partial_t u + \nabla \cdot J = f.$$

Different constitutive laws relating $J$ to $u$ lead to different PDEs.

## 0.2 Transport Equation: Motion of a Conserved Quantity

Consider first a quantity carried by a velocity field $V(t,x)$. If $u(t,x)$ is its density, then the natural flux is

$$J(t,x) = u(t,x)V(t,x).$$

Substituting this into the conservation law gives

$$\partial_t u + \nabla \cdot (uV) = f.$$

When there is no source, $f = 0$, we obtain

$$\partial_t u + \nabla \cdot (uV) = 0.$$

Using

$$\nabla \cdot (uV) = V \cdot \nabla u + u \nabla \cdot V,$$

the equation becomes

$$\partial_t u + V \cdot \nabla u + u \nabla \cdot V = 0.$$

If the velocity field is incompressible, so that $\nabla \cdot V = 0$, then

$$\partial_t u + V \cdot \nabla u = 0.$$

For a constant velocity $V = b$, this is the first-order transport equation

$$\partial_t u + b \cdot \nabla u = 0.$$

Its physical interpretation is especially transparent: the profile of $u$ is transported by the flow without changing its shape. In one dimension, for constant velocity $b$,

$$u(t,x) = u(0, x - bt).$$

Thus information moves along the straight lines $x = x_0 + bt$. These curves are the simplest examples of *characteristics*, which will be the main tool for first-order PDEs.
More generally, for

$$\partial_t u + b(t,x) \cdot \nabla u = f(t,x),$$

the characteristic curves satisfy the ordinary differential equation

$$\dot{x}(t) = b(t, x(t)).$$

Along such a trajectory,

$$\frac{d}{dt} u(t, x(t)) = \partial_t u + \dot{x}(t) \cdot \nabla u = f(t, x(t)).$$

A first-order PDE can therefore often be understood as an ODE observed along the motion of particles.

## 0.3 Heat Equation: Diffusion Driven by Temperature Gradients

The heat equation comes from the same conservation principle, but the physical law for the flux is different.

Let $u(t, x)$ denote temperature. Heat naturally flows from regions of high temperature to regions of low temperature. Fourier's law models the heat flux by

$$J = -\kappa \nabla u,$$

where $\kappa > 0$ is the thermal conductivity. The minus sign is essential: $\nabla u$ points toward increasing temperature, whereas heat flows in the opposite direction.

Suppose for simplicity that the density and specific heat of the material have been normalized to one. Conservation of thermal energy gives

$$\partial_t u + \nabla \cdot J = f.$$

Substituting Fourier's law,

$$\partial_t u - \nabla \cdot (\kappa \nabla u) = f.$$

If $\kappa$ is constant, this reduces to

$$\partial_t u - \kappa \Delta u = f.$$

After rescaling time, one often takes $\kappa = 1$, giving

$$\partial_t u - \Delta u = f.$$

In the homogeneous case,

$$\partial_t u - \Delta u = 0,$$

which is the classical heat equation.

The Laplacian

$$\Delta u = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}$$

measures, in a local sense, how the value of $u$ differs from its surroundings. If a point is hotter than its nearby environment, diffusion tends to decrease its temperature; if it is colder, diffusion tends to increase it. The heat equation therefore smooths spatial irregularities as time evolves.

This behavior is fundamentally different from transport. Transport moves information, while diffusion spreads and smooths it.

The physical problem is not determined by the PDE alone. On a bounded domain $\Omega$, one must also prescribe boundary conditions. Typical choices are

$$u = g \qquad \text{on } \partial\Omega,$$

which is a Dirichlet boundary condition, or

$$\frac{\partial u}{\partial \nu} = g \qquad \text{on } \partial\Omega,$$

which is a Neumann boundary condition. The former prescribes temperature at the boundary, while the latter prescribes the normal heat flux.

## 0.4 Laplace and Poisson Equations: Equilibrium

Suppose now that the temperature distribution has reached equilibrium. Then it no longer changes with time, so $\partial_t u = 0$. The heat equation

$$\partial_t u - \Delta u = f$$

therefore becomes

$$-\Delta u = f.$$

This is the Poisson equation.
If there is no internal source, $f = 0$, then

$$\Delta u = 0,$$

which is the Laplace equation. A function satisfying this equation is called *harmonic*.
Thus the Laplace equation may be interpreted as the stationary version of the heat equation. This simple observation already suggests why harmonic functions possess strong averaging and maximum properties: at equilibrium, there should be no isolated interior hot spot or cold spot unless the temperature is constant.
The same equation occurs far beyond heat conduction. It describes electrostatic potentials in charge-free regions, gravitational potentials away from masses, and many other equilibrium phenomena. With a source present, these models naturally lead to the Poisson equation.
This distinction between evolution and equilibrium will repeatedly appear throughout these notes:

$$\text{heat equation} \quad \longrightarrow \quad \text{Laplace/Poisson equation}$$

when the time dependence disappears.

## 0.5 Wave Equation: Newton's Law for a Vibrating String

Diffusion is not the only possible evolution. A vibrating string provides a different physical mechanism.
Consider a stretched string occupying an interval and let $u(t, x)$ denote its vertical displacement from equilibrium. Assume that the displacement and slope are small, so the motion is essentially transverse and the tension has approximately constant magnitude $T$.
Take a short piece of string between $x$ and $x+\Delta x$. The vertical components of the tension forces at its two endpoints are approximately

$$T u_x(t, x + \Delta x) - T u_x(t, x).$$

Hence the net vertical force is approximately

$$T u_{xx}(t, x)\Delta x.$$

If the string has linear mass density $\rho$, then this piece has mass approximately $\rho\Delta x$. Newton's second law gives

$$\rho\Delta x\, u_{tt} = T\Delta x\, u_{xx}.$$

After cancelling $\Delta x$,

$$u_{tt} = \frac{T}{\rho} u_{xx}.$$

Writing $c^2 = T/\rho$, we obtain

$$u_{tt} - c^2 u_{xx} = 0.$$

In several dimensions the analogous equation is

$$u_{tt} - c^2 \Delta u = 0,$$

the classical wave equation.
If an external force $f(t, x)$ acts on the medium, the equation becomes

$$u_{tt} - c^2 \Delta u = f.$$

Unlike the heat equation, the wave equation does not instantly smooth out the initial disturbance. Instead, disturbances propagate through space at a finite speed. This finite propagation speed is one of the central features of hyperbolic equations.
The state of a wave also requires two pieces of initial information:

$$u(0, x) = g(x), \qquad u_t(0, x) = h(x).$$

Physically, $g$ is the initial displacement and $h$ is the initial velocity. This reflects the second-order time derivative in the wave equation.

## 0.6 A Broader View: Fields, Systems, and Maxwell's Equations

The preceding examples involve a scalar unknown $u$, but many physical systems involve several interacting quantities. A general unknown may be a vector-valued field

$$u(t, x) = \Big(u_1(t, x), \ldots, u_m(t, x)\Big).$$

The corresponding PDE is then a system rather than a single equation.
Maxwell's equations are a fundamental example. The electric field $E$ and magnetic field $B$ depend on both space and time and satisfy a coupled system involving divergence and curl. In vacuum, these equations imply wave equations for the electromagnetic fields.
Fluid mechanics provides another major family of PDE systems. Velocity, pressure, density, and temperature may all interact. Conservation of mass, momentum, and energy leads to systems such as the Euler and Navier–Stokes equations.
These examples illustrate that PDEs are not a collection of unrelated formulas. They provide a common language for describing fields whose local behavior is constrained by physical laws.

## 0.7 Three Fundamental Types of PDE Behavior

The equations derived above exhibit three qualitatively different forms of behavior.
The transport equation

$$u_t + b \cdot \nabla u = 0$$

moves information along characteristic curves.
The heat equation

$$u_t - \Delta u = 0$$

diffuses information and smooths the solution.
The wave equation

$$u_{tt} - \Delta u = 0$$

propagates disturbances with finite speed.
Finally, the Laplace equation

$$\Delta u = 0$$

describes an equilibrium rather than a time evolution.
These differences are not merely physical interpretations. They determine which mathematical methods are useful. Characteristics are natural for first-order equations; Fourier analysis and fundamental solutions are especially effective for constant-coefficient evolution equations; maximum principles and energy estimates provide qualitative information; Green's functions represent solutions of elliptic boundary-value problems; and weak formulations allow us to discuss solutions that do not possess all derivatives in the classical sense.

## 0.8 The Role of Initial and Boundary Data

A differential equation alone rarely specifies a unique physical state. Additional information is required.
For an evolution equation, one normally prescribes initial data. For example,

$$u(0, x) = g(x)$$

specifies the initial temperature for the heat equation, while

$$u(0, x) = g(x), \qquad u_t(0, x) = h(x)$$

specifies the initial displacement and velocity for the wave equation.
On a bounded spatial domain, boundary data are also needed. The appropriate choice depends on the physical problem. Dirichlet data prescribe the value of the unknown, Neumann data prescribe its normal derivative or flux, and mixed boundary conditions combine the two.
A central question of PDE theory is therefore not simply whether one can write down a formula for $u$, but whether a problem is *well-posed*: does a solution exist, is it unique, and does it depend continuously on the prescribed data?
These questions motivate many of the estimates and principles developed later in the notes.

## 0.9 Organization of These Notes

These notes begin with first-order transport equations, where the method of characteristics gives a direct geometric description of the solution. The heat equation then introduces diffusion, Fourier analysis, fundamental solutions, Duhamel's principle, separation of variables, maximum principles, and energy estimates.
The study of elliptic equations begins with the fundamental solution of the Laplacian and Green's functions, followed by harmonic functions, mean-value properties, maximum principles, Harnack's inequality, and Perron's method. Classical differentiability is then relaxed through weak derivatives and Sobolev spaces. This leads naturally to weak solutions and variational methods, which provide an existence theory even when classical solutions are difficult to obtain.
The wave equation introduces finite propagation speed, d'Alembert's formula, spherical means, Duhamel's principle, energy methods, and the domain of dependence. Finally, Hamilton–Jacobi equations return to first-order nonlinear PDEs and connect characteristics with the calculus of variations, Lagrangian mechanics, the Legendre transform, the Hopf–Lax formula, and conservation laws.

The main theme throughout is that the form of a PDE reflects the mechanism of the physical or geometric process that produced it. Understanding that mechanism is often the first step toward understanding the equation itself.

# Chapter 1

# Preliminaries

Partial differential equations are studied in different function spaces depending on the regularity available for the solution. Classical solutions require pointwise derivatives, while weak solutions only require derivatives in an integral sense. We therefore begin with several function spaces that will be used throughout these notes.

## 1.1 Classical Function Spaces

Let $\Omega \subseteq \mathbb{R}^d$ be an open set.

**Definition 1.1.1** (Continuous Functions)**.** The space $C(\Omega)$, also denoted by $C^0(\Omega)$, is the space of continuous functions on $\Omega$:

$$C(\Omega) = \{u : \Omega \to \mathbb{R} : u \text{ is continuous}\}.$$

More generally, differentiability gives the spaces $C^k(\Omega)$.

**Definition 1.1.2** ($C^k$ Spaces)**.** For $k \in \mathbb{N}$,

$$C^k(\Omega) = \{u : \Omega \to \mathbb{R} : D^\alpha u \text{ exists and is continuous for } |\alpha| \leq k\}.$$

Thus $C^1(\Omega)$ consists of continuously differentiable functions, $C^2(\Omega)$ consists of functions with continuous derivatives up to second order, and so on.

Here a multi-index is

$$\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d, \qquad |\alpha| = \alpha_1 + \cdots + \alpha_d,$$

and

$$D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}.$$

**Definition 1.1.3** (Smooth Functions)**.** The space

$$C^\infty(\Omega) = \bigcap_{k=0}^{\infty} C^k(\Omega)$$

consists of functions having continuous derivatives of every order. Such functions are called *smooth*.

In PDEs we often need functions that vanish outside a compact subset of the domain.

**Definition 1.1.4** (Compactly Supported Smooth Functions)**.** The space $C_c^\infty(\Omega)$ consists of all smooth functions whose support is compactly contained in $\Omega$:

$$C_c^\infty(\Omega) = \{\varphi \in C^\infty(\Omega) : \operatorname{supp} \varphi \Subset \Omega\}.$$

Here

$$\operatorname{supp} \varphi = \overline{\{x \in \Omega : \varphi(x) \neq 0\}}.$$

The notation $\operatorname{supp} \varphi \Subset \Omega$ means that the support is a compact subset lying strictly inside $\Omega$.

Functions in $C_c^\infty(\Omega)$ will later be used as *test functions*. Their compact support makes integration by parts particularly convenient because no boundary term appears.

**Definition 1.1.5** (Functions Vanishing on the Boundary)**.** We write

$$C_0^k(\Omega) = \left\{ u \in C^k(\overline{\Omega}) : u = 0 \text{ on } \partial\Omega \right\}$$

when we wish to emphasize the zero boundary condition.

**Remark 1.1.6.** The notation $C_0^k(\Omega)$ is used here in the sense relevant to these notes: functions satisfying a homogeneous boundary condition. In other texts, similar notation may instead refer to the closure of compactly supported functions under a specified norm.

For functions depending on both time and space, it is useful to distinguish time derivatives from spatial derivatives.

**Definition 1.1.7** (Mixed Regularity)**.** The notation $C^{\alpha,\beta}$ indicates regularity of different orders in time and space. In these notes, $u \in C^{\alpha,\beta}$ means that $u$ possesses continuous derivatives up to order $\alpha$ in $t$ and up to order $\beta$ in $x$, as required by the context.

For example, a classical solution of the heat equation typically has one time derivative and two spatial derivatives.

## 1.2 Lebesgue Spaces

Classical spaces measure regularity pointwise. In many PDE arguments, however, it is more natural to measure the overall size of a function by integration.

**Definition 1.2.1** ($L^p$ Spaces)**.** For $1 \le p < \infty$,

$$L^p(\Omega) = \left\{ f : \Omega \to \mathbb{R} : \int_\Omega |f(x)|^p \, dx < \infty \right\},$$

where functions that agree almost everywhere are identified. The corresponding norm is

$$\|f\|_{L^p(\Omega)} = \left( \int_\Omega |f(x)|^p \, dx \right)^{1/p}.$$

The case $p = 2$ is particularly important:

$$\|f\|_{L^2(\Omega)} = \left( \int_\Omega |f(x)|^2 \, dx \right)^{1/2}.$$

It is associated with the inner product

$$\langle f, g \rangle_{L^2(\Omega)} = \int_\Omega f(x) g(x) \, dx.$$

Thus $L^2(\Omega)$ is a Hilbert space.
The limiting case $p = \infty$ is defined using the essential supremum.

**Definition 1.2.2** ($L^\infty$ Space)**.** The space $L^\infty(\Omega)$ consists of essentially bounded measurable functions, with norm

$$\|f\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_{x \in \Omega} |f(x)|.$$

The word "essential" means that sets of measure zero are ignored. Thus a function may be unbounded on a set of measure zero while still belonging to $L^\infty(\Omega)$.
For example,

$$f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0 \end{cases}$$

satisfies $f = 0$ almost everywhere and hence represents the zero element of every $L^p$ space.

**Remark 1.2.3.** The distinction between pointwise equality and equality almost everywhere becomes important when weak solutions are introduced. In $L^p$ spaces, changing a function on a set of measure zero does not change the function as an element of the space.

## 1.3 Sobolev Spaces

A classical derivative may fail to exist even when a function still has enough regularity to solve a PDE in an integral sense. Sobolev spaces are designed to capture this weaker notion of differentiability.
For now, it is enough to introduce the notation. Weak derivatives and their precise meaning will be developed when weak solutions are studied in Chapter 4.

**Definition 1.3.1** (Sobolev Space $H^k$)**.** For $k \in \mathbb{N}$, the Sobolev space $H^k(\Omega)$ consists of functions in $L^2(\Omega)$ whose weak derivatives up to order $k$ also belong to $L^2(\Omega)$:

$$H^k(\Omega) = \left\{ u \in L^2(\Omega) : D^\alpha u \in L^2(\Omega) \text{ for all } |\alpha| \leq k \right\}.$$

Its natural norm is

$$\|u\|_{H^k(\Omega)} = \left( \sum_{|\alpha| \leq k} \|D^\alpha u\|^2_{L^2(\Omega)} \right)^{1/2}.$$

In particular,

$$H^1(\Omega) = \left\{ u \in L^2(\Omega) : D_i u \in L^2(\Omega), \quad i = 1, \ldots, d \right\},$$

with

$$\|u\|^2_{H^1(\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega)}.$$

**Remark 1.3.2.** The notation

$$H^k(\Omega) = W^{k,2}(\Omega)$$

is also commonly used. More generally, $W^{k,p}(\Omega)$ is obtained by requiring the weak derivatives up to order $k$ to belong to $L^p(\Omega)$.

For homogeneous Dirichlet boundary conditions, the important space is $H^1_0(\Omega)$.

**Definition 1.3.3** (The Space $H^1_0(\Omega)$)**.** The space $H^1_0(\Omega)$ is the closure of $C^\infty_c(\Omega)$ in the $H^1(\Omega)$ norm:

$$H^1_0(\Omega) = \overline{C^\infty_c(\Omega)}^{H^1(\Omega)}.$$

Informally, $H^1_0(\Omega)$ should be viewed as the Sobolev space of functions satisfying the homogeneous boundary condition $u = 0$ on $\partial\Omega$. The rigorous interpretation of boundary values requires the trace theorem, which will not be needed explicitly in these notes.

## 1.4 Evolution Spaces

Evolution equations such as the heat and wave equations involve a function whose spatial profile changes with time. It is therefore convenient to regard

$$t \longmapsto u(t, \cdot)$$

as a curve taking values in a function space.
For example,

$$u \in L^2(0, T; H^1(\Omega))$$

means that for almost every $t \in (0, T)$, $u(t, \cdot) \in H^1(\Omega)$, and

$$\int_0^T \|u(t, \cdot)\|^2_{H^1(\Omega)} \, dt < \infty.$$

Similarly,

$$u \in C([0,T]; L^2(\Omega))$$

means that

$$t \longmapsto u(t,\cdot)$$

is continuous as a map from $[0,T]$ into $L^2(\Omega)$. In other words, if $t_n \to t$, then

$$\|u(t_n,\cdot) - u(t,\cdot)\|_{L^2(\Omega)} \to 0.$$

More generally, expressions such as

$$L^p(0,T;X) \qquad \text{and} \qquad C([0,T];X)$$

will be used when $X$ is a spatial function space.

**Remark 1.4.1.** This point of view separates temporal and spatial regularity. For example, a function may be continuous in time with values in $L^2(\Omega)$, while possessing spatial derivatives only in a weak sense. Such distinctions are essential in the study of weak solutions of evolution PDEs.

## 1.5 Regularity and the Notion of Solution

The function spaces above correspond to different meanings of the word "solution."
A *classical solution* possesses enough continuous derivatives for every term in the PDE to be evaluated pointwise. For example, a classical solution of

$$u_t - \Delta u = f$$

must have at least the time and spatial derivatives appearing in the equation.
A *weak solution* may not possess these derivatives classically. Instead, the PDE is tested against smooth functions and derivatives are transferred by integration by parts. This allows the equation to make sense under substantially weaker regularity assumptions.
Thus the hierarchy

$$C^\infty \quad \longrightarrow \quad C^k \quad \longrightarrow \quad H^k \quad \longrightarrow \quad L^2$$

should not be interpreted merely as a list of spaces. It reflects a central theme of PDE theory: when classical differentiability is unavailable, one enlarges the class of admissible functions while preserving enough structure to formulate and analyze the equation.
The precise theory of weak derivatives, Sobolev spaces, and weak solutions will be developed later when it becomes necessary.

# Chapter 2

# First-Order Transport Equations

The transport equation describes the motion of a quantity carried by a velocity field. Its central geometric idea is that the PDE becomes an ordinary differential equation when restricted to suitable curves called *characteristics*.

## 2.1 Constant-Coefficient Transport Equation

Consider the initial-value problem

$$\begin{cases} u_t + b \cdot \nabla u = f(t,x), & t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = \phi(x), & x \in \mathbb{R}^d, \end{cases}$$

where $b \in \mathbb{R}^d$ is a constant velocity vector.
We first consider the homogeneous one-dimensional equation

$$\begin{cases} u_t + cu_x = 0, & t > 0,\ x \in \mathbb{R}, \\ u(0,x) = \phi(x), & x \in \mathbb{R}, \end{cases}$$

where $c \in \mathbb{R}$ is constant.
The equation says that the value of $u$ is transported with velocity $c$. To see this, introduce

$$v(t,x) = u(t, x+ct).$$

Then

$$v_t(t,x) = u_t(t,x+ct) + cu_x(t,x+ct) = 0.$$

Hence $v$ is independent of $t$, and therefore

$$v(t,x) = v(0,x) = \phi(x).$$

Returning to $u$,

$$u(t,x) = \phi(x-ct).$$

Thus the initial profile is translated rigidly with speed $c$. If $c > 0$, it moves to the right; if $c < 0$, it moves to the left.

**Remark 2.1.1.** The transport equation does not smooth the initial data. Any jump, corner, or other irregularity in $\phi$ is simply transported along with the profile. This is fundamentally different from the heat equation.

## 2.2 Method of Characteristics

The previous calculation can be understood geometrically. Consider a curve

$$\gamma(t) = x_0 + ct.$$

Along this curve,

$$\frac{d}{dt} u(t,\gamma(t)) = u_t(t,\gamma(t)) + \gamma'(t) u_x(t,\gamma(t)).$$

Since $\gamma'(t) = c$,

$$\frac{d}{dt} u(t,\gamma(t)) = u_t + cu_x = 0.$$

Hence $u$ is constant along each characteristic:

$$u(t, x_0 + ct) = u(0, x_0) = \phi(x_0).$$

Given a point $(t,x)$, its characteristic intersects the initial line $t = 0$ at $x_0 = x - ct$. Therefore

$$u(t,x) = \phi(x-ct).$$

**Definition 2.2.1** (Characteristic Curve)**.** For the equation

$$u_t + b(t,x) \cdot \nabla u = f(t,x),$$

a characteristic curve $X(t)$ is a solution of

$$\dot{X}(t) = b(t, X(t)).$$

Along such a curve,

$$\frac{d}{dt} u(t, X(t)) = u_t(t, X(t)) + b(t, X(t)) \cdot \nabla u(t, X(t)).$$

Thus the PDE becomes

$$\frac{d}{dt} u(t, X(t)) = f(t, X(t)).$$

This is the main principle of the method of characteristics: a first-order transport PDE becomes an ODE along the correct curves.

## 2.3 Transport Equation with a Source Term

Consider

$$\begin{cases} u_t + c u_x = f(t,x), \quad t > 0,\ x \in \mathbb{R}, \\ u(0,x) = \phi(x). \end{cases}$$

Fix $(t,x)$. The characteristic passing through this point is

$$X(s) = x - c(t-s), \qquad 0 \le s \le t.$$

Indeed, $X(t) = x$ and $X'(s) = c$.
Along the characteristic,

$$\frac{d}{ds} u(s, X(s)) = f(s, X(s)).$$

Integrating from 0 to $t$,

$$u(t,x) - u(0, x-ct) = \int_0^t f\Big(s, x - c(t-s)\Big)\, ds.$$

Therefore

$$u(t,x) = \phi(x-ct) + \int_0^t f\Big(s, x - c(t-s)\Big)\, ds.$$

The two terms have clear interpretations. The first transports the initial data, while the second accumulates the source encountered along the characteristic.
In $\mathbb{R}^d$, for a constant vector $b \in \mathbb{R}^d$,

$$\begin{cases} u_t + b \cdot \nabla u = f(t,x), \\ u(0,x) = \phi(x), \end{cases}$$

the same argument gives

$$u(t,x) = \phi(x - tb) + \int_0^t f\Big(s, x - (t-s)b\Big)\, ds.$$

## 2.4 Variable Velocity Fields

Now consider the more general equation

$$\begin{cases} u_t + b(t,x) \cdot \nabla u = f(t,x), & t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = \phi(x), & x \in \mathbb{R}^d. \end{cases}$$

The characteristics are no longer straight lines. For each initial point $x_0$, let $X(t;x_0)$ solve

$$\begin{cases} \dot{X}(t;x_0) = b(t, X(t;x_0)), \\ X(0;x_0) = x_0. \end{cases}$$

Along this curve,

$$\frac{d}{dt} u(t, X(t;x_0)) = f(t, X(t;x_0)).$$

Therefore

$$u(t, X(t;x_0)) = \phi(x_0) + \int_0^t f(s, X(s;x_0))\, ds.$$

If, for each $t$, the map

$$x_0 \longmapsto X(t;x_0)$$

is invertible, then the solution can be recovered at a point $(t,x)$ by following the characteristic backward until it reaches the initial surface.

**Remark 2.4.1.** For constant $b$, the flow is simply

$$X(t;x_0) = x_0 + tb,$$

so the general characteristic method reduces to the translation formula derived earlier.

## 2.5 Nonlinear First-Order Equations

Characteristics are not restricted to linear equations. Consider a quasilinear first-order PDE of the form

$$u_t + b(t,x,u) \cdot \nabla u = f(t,x,u).$$

The characteristic system now couples the position and the value of the solution:

$$\begin{cases} \dot{X}(t) = b(t, X(t), U(t)), \\ \dot{U}(t) = f(t, X(t), U(t)), \end{cases}$$

where

$$U(t) = u(t, X(t)).$$

Thus a nonlinear PDE is transformed into a coupled system of nonlinear ODEs. The characteristics themselves may now depend on the solution.
A fundamental example is

$$u_t + u u_x = 0.$$

The characteristic equations are

$$\dot{X}(t) = U(t), \qquad \dot{U}(t) = 0.$$

Hence $U(t)$ is constant along each characteristic. If the characteristic starts from $x_0$, then

$$U(t) = \phi(x_0)$$

and

$$X(t) = x_0 + t\phi(x_0).$$

Therefore the solution is determined implicitly by

$$u(t, x) = \phi(x_0), \qquad x = x_0 + t\phi(x_0).$$

Unlike the constant-coefficient transport equation, different characteristics may now travel at different speeds because their velocity depends on $u$. This is the basic mechanism behind characteristic crossing and the formation of singularities in nonlinear first-order equations.

## 2.6 Propagation of Discontinuous Data

The handwritten examples illustrate an important feature of transport: discontinuities propagate along characteristics rather than being immediately smoothed.
Consider

$$\begin{cases} u_t + uu_x = 0, \\ u(0, x) = \phi(x). \end{cases}$$

Since $u$ is constant along a characteristic and $\dot{X} = u$, a region where $u = a$ generates characteristics of slope determined by

$$X(t) = x_0 + at.$$

For example, suppose

$$\phi(x) = \begin{cases} 0, & x \le 0, \\ 1, & x > 0. \end{cases}$$

The characteristics emerging from $x_0 < 0$ have speed 0, whereas those emerging from $x_0 > 0$ have speed 1. This leaves an expanding region between the two families of characteristics.
A continuous solution filling this region is

$$u(t, x) = \begin{cases} 0, & x \le 0, \\ \dfrac{x}{t}, & 0 < x < t, \\ 1, & x \ge t. \end{cases}$$

Indeed, in the middle region,

$$u_t = -\frac{x}{t^2}, \qquad u_x = \frac{1}{t},$$

and hence

$$u_t + uu_x = -\frac{x}{t^2} + \frac{x}{t}\frac{1}{t} = 0.$$

This solution is called a *rarefaction wave*. It illustrates that even a very simple nonlinear first-order PDE can produce behavior that does not occur for linear transport equations.

Now reverse the initial data:

$$\phi(x) = \begin{cases} 1, & x < 0, \\ 0, & x > 0. \end{cases}$$

Characteristics from the left move with speed 1, while those from the right have speed 0. They therefore intersect. After this intersection, the classical characteristic construction would assign several values of $u$ to the same point.
This indicates the breakdown of classical solutions and motivates the later introduction of weak solutions and conservation laws.

## 2.7 Summary of the Characteristic Method

For a first-order equation

$$u_t + b(t, x, u) \cdot \nabla u = f(t, x, u),$$

the method of characteristics follows three steps.

1. Solve the characteristic ODE

$$\dot{X}(t) = b(t, X(t), U(t)).$$

2. Solve the evolution of $u$ along the characteristic:

$$\dot{U}(t) = f(t, X(t), U(t)).$$

3. Use the initial data to determine the constants and reconstruct $u(t, x)$ from the characteristic curves.

For linear transport, characteristics carry information without changing its value except for the contribution of a source term. For nonlinear equations, the characteristics themselves depend on the solution and may spread apart or intersect.
This geometric viewpoint will appear again in the study of Hamilton–Jacobi equations.

## Chapter 3

# Heat Equation

Consider the heat equation

$$u_t - \Delta u = f$$

for a temperature distribution $u = u(t, x)$. As derived in the Introduction, the Laplacian describes spatial diffusion, while $f$ represents an external heat source.
This chapter develops several complementary methods for the heat equation. Fourier analysis and fundamental solutions provide explicit formulas, Duhamel's principle handles forcing terms, separation of variables is useful on bounded domains, and maximum principles and energy estimates provide qualitative information such as uniqueness.

## 3.1 Fourier Transform and the Heat Equation

We begin with the Cauchy problem

$$\begin{cases} u_t - \Delta u = 0, & t > 0,\ x \in \mathbb{R}^d, \\ u(0, x) = g(x), & x \in \mathbb{R}^d. \end{cases}$$

The Fourier transform converts differentiation in space into multiplication in frequency. This makes it particularly useful for constant-coefficient PDEs on $\mathbb{R}^d$.
For a sufficiently nice function $f : \mathbb{R}^d \to \mathbb{C}$, we use the convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\, dx.$$

The inverse Fourier transform is

$$f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{ix\cdot\xi} \widehat{f}(\xi)\, d\xi.$$

The basic differentiation rule is

$$\widehat{\partial_{x_j} f}(\xi) = i\xi_j \widehat{f}(\xi).$$

Therefore

$$\widehat{\partial_{x_j x_j} f}(\xi) = -\xi_j^2 \widehat{f}(\xi),$$

and hence

$$\widehat{\Delta f}(\xi) = -\left|\xi\right|^2 \widehat{f}(\xi).$$

Another important identity is the convolution rule. If

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y) g(y)\, dy,$$

then

$$\widehat{f * g} = \widehat{f}\,\widehat{g}.$$

These identities are the only Fourier analysis facts needed for the basic heat equation.
Taking the Fourier transform in the spatial variable $x$, the PDE becomes

$$\partial_t \widehat{u}(t, \xi) + \left|\xi\right|^2 \widehat{u}(t, \xi) = 0.$$

For each fixed $\xi$, this is an ordinary differential equation in $t$. Its solution is

$$\widehat{u}(t, \xi) = e^{-t|\xi|^2} \widehat{g}(\xi).$$

Thus solving the heat equation amounts to taking the inverse Fourier transform of the multiplier $e^{-t|\xi|^2}$.

## 3.2 Fundamental Solution

The inverse Fourier transform of $e^{-t|\xi|^2}$ is a Gaussian. This motivates the following definition.

**Definition 3.2.1** (Fundamental Solution of the Heat Equation)**.** For $t > 0$, define

$$\Phi(t, x) = \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{4t}\right).$$

The function $\Phi$ is called the *fundamental solution* or *heat kernel*.

The Fourier transform calculation gives

$$\widehat{\Phi}(t, \xi) = e^{-t|\xi|^2}.$$

Therefore

$$\widehat{u}(t, \xi) = \widehat{\Phi}(t, \xi)\widehat{g}(\xi),$$

and the convolution theorem yields

$$u(t, x) = (\Phi(t, \cdot) * g)(x).$$

Explicitly,

$$u(t, x) = \frac{1}{(4\pi t)^{d/2}} \int_{\mathbb{R}^d} \exp\left(-\frac{|x - y|^2}{4t}\right) g(y)\, dy.$$

**Theorem 3.2.2** (Solution Formula for the Heat Equation)**.** Suppose $g$ is bounded and continuous. Define

$$u(t, x) = \int_{\mathbb{R}^d} \Phi(t, x - y) g(y)\, dy.$$

Then $u$ is smooth for $t > 0$, satisfies

$$u_t - \Delta u = 0,$$

and

$$u(t, x) \to g(x) \qquad \text{as } t \to 0^+$$

at every point where $g$ is continuous.

*Proof.* For $t > 0$, the Gaussian kernel and all its spatial derivatives decay rapidly. Hence differentiation under the integral sign is allowed, and

$$u_t - \Delta u = \int_{\mathbb{R}^d} (\Phi_t - \Delta\Phi)\,(t, x - y) g(y)\, dy.$$

A direct computation shows

$$\Phi_t - \Delta\Phi = 0,$$

so $u_t - \Delta u = 0$.
Moreover,

$$\int_{\mathbb{R}^d} \Phi(t, x)\, dx = 1.$$

Thus

$$u(t, x) - g(x) = \int_{\mathbb{R}^d} \Phi(t, x - y)\Big(g(y) - g(x)\Big)\, dy.$$

As $t \to 0^+$, the mass of $\Phi(t, \cdot)$ concentrates near the origin. Continuity of $g$ at $x$ then gives

$$u(t, x) \to g(x).$$

□

The heat kernel has several useful properties:

$$\Phi(t,x) > 0, \qquad \int_{\mathbb{R}^d} \Phi(t,x)\,dx = 1,$$

and

$$\Phi(t,x) = t^{-d/2}\Phi\left(1, \frac{x}{\sqrt{t}}\right).$$

The last identity shows the natural parabolic scaling of the heat equation: space scales like $\sqrt{t}$.

**Remark 3.2.3** (Infinite Propagation Speed)**.** If $g \geq 0$ is not identically zero, then

$$u(t,x) > 0$$

for every $t > 0$ and every $x \in \mathbb{R}^d$, since $\Phi(t, x-y) > 0$.
Thus even if the initial data are compactly supported, their influence is immediately felt everywhere. The heat equation therefore has infinite propagation speed.

**Remark 3.2.4** (Smoothing Effect)**.** Even if $g$ is not smooth, convolution with the Gaussian kernel produces a smooth function for every $t > 0$. Diffusion therefore has an immediate regularizing effect.

## 3.3 Duhamel's Principle

We now consider the nonhomogeneous heat equation

$$\begin{cases} u_t - \Delta u = f(t,x), \quad t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = g(x). \end{cases}$$

The initial data are propagated by the heat kernel:

$$\int_{\mathbb{R}^d} \Phi(t, x-y) g(y)\,dy.$$

The forcing term requires a second idea.
Imagine that during a very short time interval near $s$, the source $f(s,\cdot)$ introduces a new temperature distribution. From time $s$ to time $t$, this new contribution evolves according to the homogeneous heat equation for a duration $t-s$. Its contribution at time $t$ is therefore

$$\Phi(t-s,\cdot) * f(s,\cdot).$$

Adding the contributions generated at every time $0 < s < t$ gives Duhamel's principle.

**Theorem 3.3.1** (Duhamel's Principle)**.** A solution of

$$\begin{cases} u_t - \Delta u = f(t,x), \\ u(0,x) = g(x) \end{cases}$$

is represented by

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) g(y)\,dy + \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\,dy\,ds.$$

In convolution notation,

$$u(t) = \Phi(t) * g + \int_0^t \Phi(t-s) * f(s)\, ds.$$

*Sketch of Proof.* Write

$$v(t,x) = \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\, dy\, ds.$$

Differentiating with respect to $t$,

$$v_t = f(t,x) + \int_0^t \int_{\mathbb{R}^d} \Phi_t(t-s, x-y) f(s,y)\, dy\, ds.$$

Since $\Phi_t = \Delta\Phi$,

$$v_t = f + \Delta v.$$

Hence

$$v_t - \Delta v = f.$$

Also $v(0,x) = 0$. Adding the homogeneous solution $\Phi(t) * g$ gives the desired formula. □

**Remark 3.3.2** (Fundamental Solution versus Duhamel's Principle)**.** The two methods are closely related but play different roles.
The fundamental solution describes how an initial spatial distribution is propagated:

$$g \quad \longmapsto \quad \Phi(t) * g.$$

Duhamel's principle treats a time-dependent source as a continuous superposition of new initial data introduced at different times:

$$f(s) \quad \longmapsto \quad \Phi(t-s) * f(s).$$

Thus the formulas look similar because Duhamel's principle repeatedly uses the same homogeneous solution operator. The fundamental solution concerns spatial propagation from a point source, while Duhamel's principle adds temporal superposition.

## 3.4 Separation of Variables

The Fourier transform is naturally adapted to $\mathbb{R}^d$. On a bounded domain, boundary conditions become important, and separation of variables provides another useful method. Consider the one-dimensional problem

$$\begin{cases} u_t - u_{xx} = 0, & 0 < x < L,\ t > 0, \\ u(t,0) = u(t,L) = 0, & t > 0, \\ u(0,x) = g(x), & 0 < x < L. \end{cases}$$

Assume that the solution has the form

$$u(t,x) = T(t)X(x).$$

Substituting into the PDE,

$$T'(t)X(x) - T(t)X''(x) = 0.$$

Dividing by $T(t)X(x)$,

$$\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)}.$$

The left-hand side depends only on $t$, while the right-hand side depends only on $x$. Hence both must equal a constant, say $-\lambda$:

$$\frac{T'}{T} = \frac{X''}{X} = -\lambda.$$

We obtain

$$T' + \lambda T = 0$$

and

$$\begin{cases} -X'' = \lambda X, \\ X(0) = X(L) = 0. \end{cases}$$

The spatial problem is an eigenvalue problem. Nonzero solutions exist for

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad n = 1, 2, \ldots,$$

with eigenfunctions

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

The corresponding time factors are

$$T_n(t) = e^{-\lambda_n t}.$$

Hence

$$u_n(t, x) = e^{-(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right)$$

is a solution for each $n$.
By linearity, we form

$$u(t, x) = \sum_{n=1}^{\infty} b_n e^{-(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

The initial condition requires

$$g(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

Thus $b_n$ are the Fourier sine coefficients:

$$b_n = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) \, dx.$$

Therefore

$$u(t, x) = \sum_{n=1}^{\infty} b_n e^{-(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

Each spatial frequency decays exponentially, and higher frequencies decay faster because

$$e^{-(n\pi/L)^2 t}$$

decreases rapidly as $n$ increases. This gives another explanation for the smoothing effect of the heat equation.

## 3.5 Maximum Principle

Explicit formulas are useful, but many properties of solutions can be proved without solving the equation.

Let $\Omega \subset \mathbb{R}^d$ be bounded and define the parabolic cylinder

$$\Omega_T = \Omega \times (0, T].$$

Its parabolic boundary is

$$\Gamma_T = \Big(\overline{\Omega} \times \{0\}\Big) \cup \Big(\partial\Omega \times [0, T]\Big).$$

The initial surface belongs to the parabolic boundary because the heat equation evolves forward in time.

**Theorem 3.5.1** (Weak Maximum Principle)**.** Suppose

$$u \in C^2(\Omega_T) \cap C(\overline{\Omega_T})$$

and

$$u_t - \Delta u \leq 0 \qquad \text{in } \Omega_T.$$

Then

$$\max_{\overline{\Omega_T}} u = \max_{\Gamma_T} u.$$

Similarly, if

$$u_t - \Delta u \geq 0,$$

then

$$\min_{\overline{\Omega_T}} u = \min_{\Gamma_T} u.$$

*Sketch of Proof.* First assume the strict inequality

$$u_t - \Delta u < 0.$$

Suppose $u$ attains a maximum at an interior point $(t_0, x_0)$ with $t_0 > 0$. At such a point,

$$\nabla u(t_0, x_0) = 0, \qquad \Delta u(t_0, x_0) \leq 0,$$

and, since the maximum is taken over earlier times as well,

$$u_t(t_0, x_0) \geq 0.$$

Hence

$$u_t - \Delta u \geq 0,$$

contradicting the strict inequality.

For the non-strict case, introduce

$$u_\varepsilon(t, x) = u(t, x) - \varepsilon t.$$

Then

$$(u_\varepsilon)_t - \Delta u_\varepsilon = u_t - \Delta u - \varepsilon < 0.$$

Apply the strict case to $u_\varepsilon$ and let $\varepsilon \to 0$. □

**Corollary 3.5.2** (Maximum Principle for the Heat Equation)**.** If

$$u_t - \Delta u = 0$$

in $\Omega_T$, then

$$\max_{\overline{\Omega_T}} |u| = \max_{\Gamma_T} |u| .$$

The maximum principle expresses the dissipative nature of heat flow: without an internal source, a new maximum or minimum cannot spontaneously appear in the interior.

## 3.6 Uniqueness and Stability

The maximum principle immediately gives uniqueness.

**Theorem 3.6.1** (Uniqueness)**.** Consider

$$\begin{cases} u_t - \Delta u = f, & \text{in } \Omega_T, \\ u = g, & \text{on } \Gamma_T. \end{cases}$$

There is at most one classical solution.

*Proof.* Suppose $u_1$ and $u_2$ are two solutions and let

$$w = u_1 - u_2.$$

Then

$$w_t - \Delta w = 0$$

in $\Omega_T$, and $w = 0$ on $\Gamma_T$.
By the maximum principle,

$$\max_{\overline{\Omega_T}} w \leq 0.$$

Applying the same argument to $-w$,

$$\max_{\overline{\Omega_T}} (-w) \leq 0.$$

Hence $w = 0$, so $u_1 = u_2$. □

The same argument gives continuous dependence on boundary and initial data.

**Proposition 3.6.2** (Stability)**.** Suppose $u_1$ and $u_2$ satisfy the same heat equation but have parabolic boundary data $g_1$ and $g_2$. Then

$$\|u_1 - u_2\|_{L^\infty(\Omega_T)} \leq \|g_1 - g_2\|_{L^\infty(\Gamma_T)}.$$

*Proof.* Set $w = u_1 - u_2$. Then

$$w_t - \Delta w = 0.$$

Applying the maximum principle to $w$ and $-w$ gives

$$|w(t,x)| \leq \max_{\Gamma_T} |w|.$$

The result follows. □

## 3.7 Energy Method

A second approach to uniqueness is the energy method. Unlike the maximum principle, this method extends naturally to many PDEs for which no maximum principle is available. Suppose

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \Omega_T, \\ u = 0, & \text{on } \partial\Omega \times (0,T], \\ u(0,x) = g(x). \end{cases}$$

Define the energy

$$E(t) = \frac{1}{2} \int_\Omega |u(t,x)|^2 \, dx.$$

Then

$$E'(t) = \int_\Omega u u_t \, dx.$$

Since $u_t = \Delta u$,

$$E'(t) = \int_\Omega u \Delta u \, dx.$$

Integration by parts gives

$$\int_\Omega u \Delta u \, dx = -\int_\Omega |\nabla u|^2 \, dx + \int_{\partial\Omega} u \frac{\partial u}{\partial \nu} \, dS.$$

Since $u = 0$ on $\partial\Omega$, the boundary term vanishes. Hence

$$E'(t) = -\int_\Omega |\nabla u|^2 \, dx \leq 0.$$

Therefore the $L^2$-energy is nonincreasing:

$$E(t) \leq E(0).$$

Equivalently,

$$\|u(t,\cdot)\|_{L^2(\Omega)} \leq \|g\|_{L^2(\Omega)}.$$

This is another manifestation of diffusion: the heat equation dissipates energy over time.

**Theorem 3.7.1** (Uniqueness by the Energy Method)**.** Suppose $u_1$ and $u_2$ solve the same homogeneous Dirichlet initial-boundary value problem for the heat equation. Then $u_1 = u_2$.

*Proof.* Let

$$w = u_1 - u_2.$$

Then

$$\begin{cases} w_t - \Delta w = 0, \\ w = 0 \quad \text{on } \partial\Omega \times (0,T], \\ w(0,x) = 0. \end{cases}$$

Define

$$E(t) = \frac{1}{2} \int_\Omega |w(t,x)|^2 \, dx.$$

The previous calculation gives $E'(t) \leq 0$. Since $E(0) = 0$ and $E(t) \geq 0$,

$$E(t) = 0$$

for every $t$. Hence $w = 0$, and therefore $u_1 = u_2$. □

**Remark 3.7.2.** The maximum principle controls the solution in the $L^\infty$ sense, whereas the energy method controls it in the $L^2$ sense. They are different techniques proving related qualitative properties.
The energy method is especially important because its basic ingredients, multiplication by the solution and integration by parts, continue to make sense in weak formulations. This will become important when weak solutions are introduced later.

## 3.8 Summary

The heat equation can be approached from several complementary directions.
On the whole space, Fourier transform converts

$$u_t - \Delta u = 0$$

into a family of ordinary differential equations in frequency space. The inverse transform produces the Gaussian heat kernel

$$\Phi(t, x) = \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{4t}\right),$$

which propagates initial data by convolution.
For a nonhomogeneous equation, Duhamel's principle superposes the effects of the source at different times. On bounded domains, separation of variables replaces the continuous Fourier transform by an expansion in eigenfunctions.
Finally, the maximum principle and energy method allow us to obtain uniqueness and stability without explicitly solving the equation. These methods will repeatedly reappear for other PDEs.

# Chapter 4

# Elliptic Equations

The basic elliptic equation considered in this chapter is the Poisson equation

$$-\Delta u = f \qquad \text{in } \Omega.$$

When $f = 0$, it reduces to the Laplace equation

$$\Delta u = 0.$$

Unlike the heat and wave equations, these equations contain no time variable. They typically describe equilibrium states. We first study classical solutions through fundamental solutions, Green's functions, and properties of harmonic functions. We then introduce weak solutions and variational methods, which allow us to work under much weaker regularity assumptions.

## 4.1 Fundamental Solution of the Laplace Equation

We first seek a function $\Phi$ satisfying

$$-\Delta \Phi = \delta_0$$

in the distributional sense, where $\delta_0$ denotes the Dirac mass at the origin.
Away from the origin, this requires

$$\Delta \Phi = 0 \qquad \text{in } \mathbb{R}^d \setminus \{0\}.$$

Because the Laplacian is rotationally invariant, we look for a radial solution

$$\Phi(x) = \phi(r), \qquad r = |x|\,.$$

For a radial function,

$$\Delta \Phi = \phi''(r) + \frac{d-1}{r}\phi'(r).$$

Hence

$$\phi''(r) + \frac{d-1}{r}\phi'(r) = 0.$$

Equivalently,

$$\frac{d}{dr}\left(r^{d-1}\phi'(r)\right) = 0.$$

Therefore

$$\phi'(r) = Cr^{1-d}.$$

For $d \geq 3$,

$$\phi(r) = C_1 r^{2-d} + C_2,$$

while for $d = 2$,

$$\phi(r) = C_1 \log r + C_2.$$

After choosing the normalization so that $-\Delta\Phi = \delta_0$, we obtain the fundamental solution.

**Definition 4.1.1** (Fundamental Solution of the Laplacian)**.** Let $\omega_d = |B_1(0)|$ be the volume of the unit ball in $\mathbb{R}^d$. The fundamental solution of $-\Delta$ is

$$\Phi(x) = \begin{cases} \dfrac{1}{d(d-2)\omega_d}\dfrac{1}{|x|^{d-2}}, & d \geq 3, \\[2ex] -\dfrac{1}{2\pi}\log|x|\,, & d = 2. \end{cases}$$

It satisfies

$$-\Delta\Phi = \delta_0$$

in the distributional sense.

The normalization can be understood by integrating over a ball. For $d \geq 3$,

$$\Phi(x) = C\,|x|^{2-d}.$$

Then

$$\frac{\partial\Phi}{\partial\nu} = C(2-d)r^{1-d}$$

on $\partial B_r(0)$. By the divergence theorem,

$$-\int_{\partial B_r(0)} \frac{\partial\Phi}{\partial\nu}\,dS = -C(2-d)r^{1-d}\,|\partial B_r(0)|\,.$$

Since

$$|\partial B_r(0)| = d\omega_d r^{d-1},$$

choosing

$$C = \frac{1}{d(d-2)\omega_d}$$

makes the total flux equal to one.
Now consider

$$-\Delta u = f \qquad \text{in } \mathbb{R}^d.$$

Formally,

$$u = \Phi * f,$$

that is,

$$u(x) = \int_{\mathbb{R}^d} \Phi(x-y)f(y)\,dy.$$

Indeed,

$$-\Delta u = (-\Delta\Phi) * f = \delta_0 * f = f.$$

Thus the fundamental solution converts the Poisson equation on the whole space into an integral formula.

## 4.2 Green's Functions and Representation Formula

The fundamental solution solves the whole-space problem, but on a bounded domain it generally does not satisfy the prescribed boundary condition.
Consider the Dirichlet problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

To incorporate the boundary condition, we modify the fundamental solution.

**Definition 4.2.1** (Green's Function)**.** For fixed $x \in \Omega$, the Green's function is

$$G(x,y) = \Phi(x-y) - \phi^x(y),$$

where $\phi^x$ is harmonic in $\Omega$ and satisfies

$$\phi^x(y) = \Phi(x-y) \qquad \text{on } \partial\Omega.$$

Consequently,

$$\begin{cases} -\Delta_y G(x,y) = \delta_x(y), & y \in \Omega, \\ G(x,y) = 0, & y \in \partial\Omega. \end{cases}$$

The fundamental solution contains the singularity required to reproduce the source, while the harmonic correction is chosen precisely to enforce the boundary condition.
The basic identity behind the representation formula is Green's second identity:

$$\int_\Omega (u\Delta v - v\Delta u)\, dy = \int_{\partial\Omega} \left( u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu} \right) dS.$$

Taking $v(y) = G(x,y)$, we obtain the following representation.

**Theorem 4.2.2** (Green's Representation Formula)**.** Suppose

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

Then, formally,

$$u(x) = \int_\Omega G(x,y) f(y)\, dy - \int_{\partial\Omega} g(y) \frac{\partial G}{\partial \nu_y}(x,y)\, dS_y.$$

The first term describes the effect of the source inside the domain. The second term incorporates the prescribed boundary data.

**Remark 4.2.3.** The fundamental solution and Green's function solve closely related but different problems. The fundamental solution is adapted to the whole space, while Green's function modifies it to respect the geometry and boundary condition of a particular domain.

## 4.3 Harmonic Functions and the Mean Value Property

**Definition 4.3.1** (Harmonic Function)**.** A function $u \in C^2(\Omega)$ is called *harmonic* if

$$\Delta u = 0 \qquad \text{in } \Omega.$$

Harmonic functions satisfy a remarkable averaging property.

**Theorem 4.3.2** (Mean Value Property)**.** Suppose $u$ is harmonic in $\Omega$. If

$$\overline{B_r(x)} \subset \Omega,$$

then

$$u(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y)\, dS_y.$$

Equivalently,

$$u(x) = \frac{1}{|B_r(x)|} \int_{B_r(x)} u(y)\, dy.$$

*Sketch of Proof.* Fix $x \in \Omega$ and define the spherical average

$$\varphi(r) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y)\, dS_y.$$

After writing $y = x + rz$, this becomes

$$\varphi(r) = \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} u(x + rz)\, dS_z.$$

Differentiating,

$$\varphi'(r) = \frac{1}{|\partial B_1(0)|} \int_{\partial B_1(0)} \nabla u(x + rz) \cdot z\, dS_z.$$

Rescaling back to $\partial B_r(x)$,

$$\varphi'(r) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} \frac{\partial u}{\partial \nu}\, dS.$$

By the divergence theorem,

$$\int_{\partial B_r(x)} \frac{\partial u}{\partial \nu}\, dS = \int_{B_r(x)} \Delta u\, dy = 0.$$

Hence $\varphi'(r) = 0$, so $\varphi(r)$ is constant. Taking $r \to 0$,

$$\varphi(r) \to u(x).$$

Therefore the spherical mean equals $u(x)$.
The ball-average formula follows by integrating the spherical mean in the radial variable. □

**Remark 4.3.3.** The mean value property gives a physical interpretation of equilibrium: the value of a harmonic function at a point is exactly balanced by the values surrounding it.

The converse is also true.

**Theorem 4.3.4** (Converse Mean Value Property)**.** If $u \in C(\Omega)$ satisfies the mean value property on every ball compactly contained in $\Omega$, then $u$ is harmonic.

## 4.4 Maximum Principle for Harmonic Functions

The mean value property immediately implies a maximum principle.

**Theorem 4.4.1** (Strong Maximum Principle)**.** Let $\Omega$ be connected and let $u \in C^2(\Omega)$ be harmonic. If $u$ attains its maximum at an interior point of $\Omega$, then $u$ is constant in $\Omega$.

*Proof.* Suppose $u$ attains its maximum $M$ at $x_0 \in \Omega$. For any ball $\overline{B_r(x_0)} \subset \Omega$, the mean value property gives

$$M = u(x_0) = \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} u(y)\, dy.$$

Since $u(y) \leq M$, the average can equal $M$ only if

$$u(y) = M$$

throughout the ball. Thus the set on which $u = M$ is both open and closed in the connected domain $\Omega$. Hence $u \equiv M$. □

**Corollary 4.4.2** (Weak Maximum Principle)**.** If $\Omega$ is bounded and

$$u \in C^2(\Omega) \cap C(\overline{\Omega})$$

is harmonic, then

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u,$$

and

$$\min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

**Corollary 4.4.3** (Uniqueness of the Dirichlet Problem)**.** The Dirichlet problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = g, & \text{on } \partial\Omega \end{cases}$$

has at most one classical solution.

*Proof.* Suppose $u_1$ and $u_2$ are two solutions and set

$$w = u_1 - u_2.$$

Then

$$\Delta w = 0 \qquad \text{in } \Omega$$

and $w = 0$ on $\partial\Omega$. By the maximum principle,

$$\max_{\overline{\Omega}} w \leq 0.$$

Applying the same argument to $-w$ gives $w \geq 0$. Hence $w = 0$, so $u_1 = u_2$. □

## 4.5 Harnack's Inequality

The maximum principle gives qualitative information about harmonic functions. Harnack's inequality gives a quantitative comparison between their values.

**Theorem 4.5.1** (Harnack's Inequality)**.** Let $u \geq 0$ be harmonic in $\Omega$. If $V \Subset \Omega$ is connected, then there exists a constant $C > 0$, depending only on $V$ and $\Omega$, such that

$$\sup_V u \leq C \inf_V u.$$

*Sketch of Proof.* First consider two nearby points contained in a ball

$$B_{2r}(x_0) \Subset \Omega.$$

The Poisson representation formula for harmonic functions expresses $u(x)$ as a weighted average of its boundary values. For points lying well inside the ball, the corresponding Poisson kernels are comparable. Hence

$$u(x) \leq Cu(y)$$

for $x, y$ in a smaller concentric ball.
Since $V \Subset \Omega$ is connected, any two points of $V$ can be joined by a finite chain of overlapping balls staying inside $\Omega$. Applying the local estimate successively along this chain gives

$$\sup_V u \leq C \inf_V u.$$

□

**Corollary 4.5.2.** If $u \geq 0$ is harmonic in a connected domain and $u(x_0) = 0$ at some interior point, then

$$u \equiv 0.$$

Harnack's inequality therefore shows that a nonnegative harmonic function cannot be very small at one interior point and arbitrarily large at another nearby point.

## 4.6 Perron's Method

The maximum principle gives uniqueness for the Dirichlet problem, but it does not by itself prove existence. Perron's method constructs a harmonic solution from subharmonic functions.

**Definition 4.6.1** (Subharmonic Function)**.** A function $v \in C^2(\Omega)$ is called *subharmonic* if

$$\Delta v \geq 0 \qquad \text{in } \Omega.$$

Similarly, $v$ is *superharmonic* if

$$\Delta v \leq 0.$$

Consider the Dirichlet problem

$$\begin{cases} \Delta u = 0, & \text{in } \Omega, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

Define the Perron class

$$\mathcal{S} = \{v : v \text{ is subharmonic in } \Omega, \quad v \leq g \text{ on } \partial\Omega\}.$$

The candidate solution is

$$u(x) = \sup_{v \in \mathcal{S}} v(x).$$

The central idea is that taking the largest admissible subharmonic function forces the resulting function to become harmonic.

**Theorem 4.6.2** (Perron's Method)**.** Under suitable regularity assumptions on the domain and boundary data, the Perron function

$$u(x) = \sup_{v \in \mathcal{S}} v(x)$$

is harmonic in $\Omega$ and attains the prescribed boundary data.

*Idea of Proof.* The proof consists of two main steps.
First, one performs a *harmonic replacement*: inside a ball $B \Subset \Omega$, replace a subharmonic function by the harmonic function having the same values on $\partial B$. The maximum principle shows that the replacement is no smaller than the original function and remains admissible.
Second, apply such replacements to a sequence of functions whose values at a fixed point approach the supremum defining $u$. Interior estimates and compactness allow one to extract a harmonic limit. By maximality, this limit must coincide with the Perron function. The final boundary behavior is obtained using suitable barrier functions. □

**Remark 4.6.3.** Perron's method is an existence method based primarily on comparison and the maximum principle. Later we will see a very different existence method: instead of taking a supremum of subsolutions, we minimize an energy functional.

## 4.7 Why Weak Solutions Are Needed

So far, a solution of

$$-\Delta u = f$$

has been assumed to possess second derivatives. This is natural when an explicit smooth formula is available, but it is often too restrictive for general boundary-value problems.
There is another way to read the equation. Suppose temporarily that $u$ is smooth and satisfies

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Take any test function $\varphi \in C_c^\infty(\Omega)$. Multiplying the equation by $\varphi$ and integrating,

$$\int_\Omega (-\Delta u)\varphi \, dx = \int_\Omega f\varphi \, dx.$$

Integration by parts gives

$$\int_\Omega \nabla u \cdot \nabla \varphi \, dx = \int_\Omega f\varphi \, dx.$$

The important observation is that the final identity involves only first derivatives of $u$. It therefore continues to make sense even when $\Delta u$ does not exist pointwise.
This motivates the idea of a weak solution: instead of requiring the PDE to hold at every point, we require its integrated form to hold against every smooth test function.
To make this precise, we first need a weaker notion of derivative.

## 4.8 Weak Derivatives

For a smooth function $u$ and $\varphi \in C_c^\infty(\Omega)$, integration by parts gives

$$\int_\Omega u \, \partial_i \varphi \, dx = -\int_\Omega (\partial_i u)\varphi \, dx.$$

The boundary term disappears because $\varphi$ has compact support.
The key idea is to reverse this identity and use it as the definition of a derivative.

**Definition 4.8.1** (Weak Derivative)**.** Let $u \in L^1_{\mathrm{loc}}(\Omega)$. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is called the $i$-th weak derivative of $u$ if

$$\int_\Omega u \, \partial_i \varphi \, dx = -\int_\Omega v\varphi \, dx$$

for every $\varphi \in C_c^\infty(\Omega)$.
We then write

$$v = D_i u.$$

More generally, $v = D^\alpha u$ is the weak derivative of order $\alpha$ if

$$\int_\Omega u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v\varphi \, dx$$

for every $\varphi \in C_c^\infty(\Omega)$.

**Example 4.8.2.** Consider

$$u(x) = |x|$$

on $(-1,1)$. The classical derivative does not exist at $x=0$, but define

$$v(x)=\begin{cases}-1, & x<0,\\ 1, & x>0.\end{cases}$$

For every $\varphi\in C_c^\infty(-1,1)$,

$$\int_{-1}^{1}|x|\,\varphi'(x)\,dx=-\int_{-1}^{1}v(x)\varphi(x)\,dx.$$

Hence

$$Du=\operatorname{sgn}(x)$$

in the weak sense.

Thus weak differentiability allows isolated failures of classical differentiability without losing the derivative altogether.

**Proposition 4.8.3** (Uniqueness of Weak Derivatives)**.** If a weak derivative of $u$ exists, then it is unique up to equality almost everywhere.

## 4.9 Sobolev Spaces

Weak derivatives become useful when they are combined with integrability.

**Definition 4.9.1** (Sobolev Space)**.** For $k\in\mathbb{N}$,

$$H^k(\Omega)=\left\{u\in L^2(\Omega):D^\alpha u\in L^2(\Omega)\text{ for all }|\alpha|\le k\right\}.$$

The norm is

$$\|u\|_{H^k(\Omega)}^2=\sum_{|\alpha|\le k}\|D^\alpha u\|_{L^2(\Omega)}^2.$$

In particular,

$$H^1(\Omega)=\left\{u\in L^2(\Omega):\nabla u\in L^2(\Omega)\right\},$$

with

$$\|u\|_{H^1(\Omega)}^2=\|u\|_{L^2(\Omega)}^2+\|\nabla u\|_{L^2(\Omega)}^2.$$

As introduced in Chapter 1,

$$H_0^1(\Omega)=\overline{C_c^\infty(\Omega)}^{H^1(\Omega)}.$$

This is the natural space for homogeneous Dirichlet boundary conditions.

**Remark 4.9.2.** The passage

$$C^1(\Omega)\quad\longrightarrow\quad H^1(\Omega)$$

allows functions whose derivatives exist only weakly. The advantage is that $H^1(\Omega)$ is a Hilbert space, so geometric and compactness arguments from functional analysis become available.

## 4.10 Weak Formulation of the Poisson Equation

Consider

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

The preceding integration-by-parts calculation motivates the following definition.

**Definition 4.10.1** (Weak Solution)**.** Let $f \in L^2(\Omega)$. A function $u \in H_0^1(\Omega)$ is called a weak solution of the Poisson equation if

$$\int_\Omega \nabla u \cdot \nabla \varphi \, dx = \int_\Omega f \varphi \, dx$$

for every $\varphi \in H_0^1(\Omega)$.

Notice what has changed. The classical equation

$$-\Delta u = f$$

requires two derivatives of $u$, while the weak formulation only requires

$$u \in H_0^1(\Omega).$$

One derivative has effectively been transferred from $u$ to the test function by integration by parts.
Every sufficiently smooth classical solution is therefore a weak solution. The converse is not automatic: recovering classical differentiability from a weak solution is a separate question called *regularity theory*.

## 4.11 Poincaré Inequality

A basic estimate on $H_0^1(\Omega)$ is the Poincaré inequality.

**Theorem 4.11.1** (Poincaré Inequality)**.** Let $\Omega \subset \mathbb{R}^d$ be bounded. There exists a constant $C = C(\Omega) > 0$ such that

$$\|u\|_{L^2(\Omega)} \leq C \, \|\nabla u\|_{L^2(\Omega)}$$

for every $u \in H_0^1(\Omega)$.

Thus, on $H_0^1(\Omega)$, the gradient controls the entire function.
In particular,

$$\|u\|_{H^1(\Omega)}^2 = \|u\|_{L^2(\Omega)}^2 + \|\nabla u\|_{L^2(\Omega)}^2 \leq (C^2 + 1) \, \|\nabla u\|_{L^2(\Omega)}^2 .$$

Therefore

$$\|\nabla u\|_{L^2(\Omega)}$$

is an equivalent norm on $H_0^1(\Omega)$.

**Remark 4.11.2.** The zero boundary condition is essential. Without it, a nonzero constant function would satisfy $\nabla u = 0$ while $\|u\|_{L^2(\Omega)} > 0$.

## 4.12 Energy Functional and Dirichlet Principle

There is a second way to arrive at the Poisson equation.
Define the energy functional

$$I[u] = \frac{1}{2}\int_\Omega |\nabla u|^2 \, dx - \int_\Omega f u \, dx.$$

Suppose $u$ minimizes $I$ over $H_0^1(\Omega)$. For any $\varphi \in H_0^1(\Omega)$, consider the variation

$$u_\varepsilon = u + \varepsilon\varphi.$$

Since $u$ is a minimizer,

$$\frac{d}{d\varepsilon} I[u+\varepsilon\varphi]\Big|_{\varepsilon=0} = 0.$$

Now

$$I[u+\varepsilon\varphi] = \frac{1}{2}\int_\Omega |\nabla u + \varepsilon\nabla\varphi|^2 \, dx - \int_\Omega f(u+\varepsilon\varphi)\, dx.$$

Differentiating at $\varepsilon = 0$,

$$\int_\Omega \nabla u \cdot \nabla\varphi \, dx - \int_\Omega f\varphi \, dx = 0.$$

Hence

$$\int_\Omega \nabla u \cdot \nabla\varphi \, dx = \int_\Omega f\varphi \, dx.$$

But this is exactly the weak formulation of

$$-\Delta u = f.$$

**Theorem 4.12.1** (Dirichlet Principle)**.** A minimizer of

$$I[v] = \frac{1}{2}\int_\Omega |\nabla v|^2 \, dx - \int_\Omega f v \, dx$$

over $H_0^1(\Omega)$ is a weak solution of

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

This provides an important connection:

$$\text{PDE} \quad \longleftrightarrow \quad \text{weak formulation} \quad \longleftrightarrow \quad \text{energy minimization}.$$

## 4.13 Existence of a Weak Solution

We now explain why the minimization problem actually has a solution.
The main idea is the *direct method of the calculus of variations*: take a minimizing sequence, prove that it cannot escape to infinity, extract a weakly convergent subsequence, and show that the limit is still a minimizer.
Let

$$m = \inf_{v \in H_0^1(\Omega)} I[v].$$

Choose a minimizing sequence

$$u_k \in H_0^1(\Omega)$$

such that

$$I[u_k] \to m.$$

We first need to show that $\{u_k\}$ is bounded in $H_0^1(\Omega)$.
By the Cauchy–Schwarz inequality,

$$\left| \int_\Omega f u_k \, dx \right| \leq \|f\|_{L^2(\Omega)} \, \|u_k\|_{L^2(\Omega)} .$$

By the Poincaré inequality,

$$\|u_k\|_{L^2(\Omega)} \leq C \, \|\nabla u_k\|_{L^2(\Omega)} .$$

Hence

$$I[u_k] \geq \frac{1}{2} \, \|\nabla u_k\|^2_{L^2(\Omega)} - C \, \|f\|_{L^2(\Omega)} \, \|\nabla u_k\|_{L^2(\Omega)} .$$

The quadratic term dominates for large $\|\nabla u_k\|_{L^2(\Omega)}$. Since $I[u_k]$ remains bounded, the sequence

$$\|\nabla u_k\|_{L^2(\Omega)}$$

must be bounded. Poincaré's inequality then implies that $\{u_k\}$ is bounded in $H_0^1(\Omega)$.
Since $H_0^1(\Omega)$ is a Hilbert space, a bounded sequence has a weakly convergent subsequence.
Thus, after passing to a subsequence,

$$u_k \rightharpoonup u \qquad \text{weakly in } H_0^1(\Omega).$$

Weak convergence means that

$$\int_\Omega u_k \varphi \, dx \to \int_\Omega u \varphi \, dx$$

and

$$\int_\Omega \nabla u_k \cdot \nabla \varphi \, dx \to \int_\Omega \nabla u \cdot \nabla \varphi \, dx$$

for suitable test functions $\varphi$.
The norm is weakly lower semicontinuous:

$$\|\nabla u\|^2_{L^2(\Omega)} \leq \liminf_{k \to \infty} \|\nabla u_k\|^2_{L^2(\Omega)} .$$

At the same time,

$$\int_\Omega f u_k \, dx \to \int_\Omega f u \, dx.$$

Therefore

$$I[u] \leq \liminf_{k \to \infty} I[u_k] = m.$$

Since $m$ is the infimum,

$$I[u] = m.$$

Thus $u$ is a minimizer.
By the Dirichlet principle, $u$ is a weak solution.

**Theorem 4.13.1** (Existence of Weak Solutions)**.** Let $\Omega \subset \mathbb{R}^d$ be bounded and let $f \in L^2(\Omega)$. Then the problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega \end{cases}$$

has a weak solution $u \in H_0^1(\Omega)$.

## 4.14 Uniqueness of the Weak Solution

The weak formulation also makes uniqueness straightforward.

**Theorem 4.14.1** (Uniqueness of Weak Solutions)**.** The weak solution of

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega \end{cases}$$

is unique.

*Proof.* Suppose $u_1, u_2 \in H_0^1(\Omega)$ are weak solutions and set

$$w = u_1 - u_2.$$

Then

$$\int_\Omega \nabla w \cdot \nabla \varphi \, dx = 0$$

for every $\varphi \in H_0^1(\Omega)$.
Since $w \in H_0^1(\Omega)$, we may choose $\varphi = w$. Then

$$\int_\Omega |\nabla w|^2 \, dx = 0.$$

Hence

$$\nabla w = 0$$

almost everywhere. By the Poincaré inequality,

$$\|w\|_{L^2(\Omega)} \leq C \, \|\nabla w\|_{L^2(\Omega)} = 0.$$

Therefore $w = 0$ almost everywhere, and $u_1 = u_2$. □

This argument is the weak analogue of the energy method used for the heat equation.

## 4.15 The Logic of Weak Solutions

The weak-solution framework can be summarized by the following chain:

$$-\Delta u = f \quad \longrightarrow \quad \int_\Omega \nabla u \cdot \nabla \varphi \, dx = \int_\Omega f \varphi \, dx \quad \longrightarrow \quad u \in H_0^1(\Omega).$$

The first step uses integration by parts to reduce the number of derivatives required of $u$. Weak derivatives then provide the correct notion of differentiability, and Sobolev spaces provide the natural setting in which the resulting integral identity makes sense.
The same identity also arises as the Euler–Lagrange condition for the energy

$$I[u] = \frac{1}{2} \int_\Omega |\nabla u|^2 \, dx - \int_\Omega f u \, dx.$$

This converts the existence problem for a PDE into a minimization problem.
The Poincaré inequality prevents minimizing sequences from becoming unbounded, weak compactness produces a candidate limit, and weak lower semicontinuity ensures that the limit minimizes the energy. Finally, testing the weak equation with the solution itself gives uniqueness.
Thus the main structure is

classical PDE $\longrightarrow$ weak formulation $\longrightarrow$ Sobolev space $\longrightarrow$ energy minimization $\longrightarrow$ existence and uni

This viewpoint is one of the central ideas in modern PDE theory.

# Chapter 5

# Wave Equation

The wave equation models the propagation of disturbances such as vibrating strings, acoustic waves, and electromagnetic waves. Its standard form is

$$u_{tt} - \Delta u = f(t, x),$$

with initial conditions

$$u(0, x) = g(x), \qquad u_t(0, x) = h(x).$$

Unlike the heat equation, the wave equation has finite propagation speed: information travels through space at a definite speed rather than being felt instantaneously everywhere. For convenience, define the d'Alembert operator

$$\Box = \partial_{tt} - \Delta.$$

## 5.1 One-Dimensional Wave Equation and d'Alembert Formula

Consider the one-dimensional Cauchy problem

$$\begin{cases} u_{tt} - u_{xx} = 0, & t > 0,\ x \in \mathbb{R}, \\ u(0, x) = g(x), & x \in \mathbb{R}, \\ u_t(0, x) = h(x), & x \in \mathbb{R}. \end{cases}$$

The solution is given explicitly by d'Alembert's formula.

**Theorem 5.1.1** (d'Alembert Formula)**.** Suppose $g \in C^2(\mathbb{R})$ and $h \in C^1(\mathbb{R})$. Then

$$u(t, x) = \frac{1}{2}\big[g(x - t) + g(x + t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy$$

solves the one-dimensional wave equation with the prescribed initial conditions.

There are several ways to derive this formula.

## 5.2 Derivation by Fourier Transform

Take the Fourier transform in the spatial variable $x$. Using

$$\widehat{u_{xx}}(t, \xi) = -\xi^2 \widehat{u}(t, \xi),$$

the equation becomes

$$\widehat{u}_{tt} + \xi^2 \widehat{u} = 0.$$

For each fixed $\xi$, this is a harmonic oscillator. The initial conditions are

$$\widehat{u}(0, \xi) = \widehat{g}(\xi), \qquad \widehat{u}_t(0, \xi) = \widehat{h}(\xi).$$

Therefore

$$\widehat{u}(t, \xi) = \widehat{g}(\xi)\cos(t\,|\xi|) + \widehat{h}(\xi)\frac{\sin(t\,|\xi|)}{|\xi|}.$$

The inverse Fourier transform of the first multiplier gives

$$\frac{1}{2}\big[g(x - t) + g(x + t)\big],$$

while the second gives

$$\frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy.$$

Hence

$$u(t, x) = \frac{1}{2}\big[g(x - t) + g(x + t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy.$$

The Fourier viewpoint interprets each frequency as an independent harmonic oscillator.

## 5.3 Derivation by Characteristics

The one-dimensional wave operator factors as

$$\partial_{tt} - \partial_{xx} = (\partial_t - \partial_x)(\partial_t + \partial_x).$$

Thus

$$(\partial_t - \partial_x)(\partial_t + \partial_x)u = 0.$$

Set

$$v = (\partial_t + \partial_x)u.$$

Then

$$v_t - v_x = 0.$$

This is a transport equation, so $v$ is constant along the appropriate characteristics. Equivalently, the general solution of the homogeneous wave equation has the form

$$u(t, x) = F(x + t) + G(x - t).$$

The initial condition gives

$$F(x) + G(x) = g(x),$$

and differentiating,

$$F'(x) - G'(x) = h(x).$$

Solving for $F$ and $G$,

$$F(x) = \frac{1}{2}g(x) + \frac{1}{2}\int_0^x h(y)\,dy,$$

and

$$G(x) = \frac{1}{2}g(x) - \frac{1}{2}\int_0^x h(y)\,dy.$$

Substitution again gives d'Alembert's formula.

**Remark 5.3.1.** The two derivations emphasize different structures. Fourier transform decomposes the solution into spatial frequencies, while the factorization of the wave operator reveals the two characteristic directions

$$x + t = \text{constant}, \qquad x - t = \text{constant}.$$

## 5.4 Finite Propagation Speed

d'Alembert's formula immediately reveals an important property.
The value $u(t, x)$ depends only on

$$g(x - t), \qquad g(x + t),$$

and the values of $h$ in

$$[x - t, x + t].$$

Thus the solution at $(t, x)$ depends only on initial data lying within distance $t$ of $x$.

**Proposition 5.4.1** (Finite Propagation Speed)**.** Suppose

$$g(y) = h(y) = 0$$

for every $y$ satisfying

$$|y - x_0| \le t_0.$$

Then

$$u(t, x) = 0$$

whenever

$$0 \le t \le t_0, \qquad |x - x_0| \le t_0 - t.$$

This region is the backward cone with vertex $(t_0, x_0)$.

**Remark 5.4.2.** This behavior contrasts sharply with the heat equation. A localized disturbance in the heat equation immediately influences the entire space, whereas a wave disturbance travels with finite speed.

More generally,

$$u_{tt} - a^2 \Delta u = 0$$

has propagation speed $a > 0$.

## 5.5 A Boundary Problem on the Half-Line

Consider

$$\begin{cases} u_{tt} - u_{xx} = 0, & t > 0,\ x > 0, \\ u(0, x) = g(x), & x > 0, \\ u_t(0, x) = h(x), & x > 0, \\ u(t, 0) = 0, & t > 0. \end{cases}$$

The homogeneous Dirichlet boundary condition can be incorporated by odd extension. Define

$$\widetilde{g}(x) = \begin{cases} g(x), & x > 0, \\ -g(-x), & x < 0, \end{cases}$$

and similarly

$$\widetilde{h}(x) = \begin{cases} h(x), & x > 0, \\ -h(-x), & x < 0. \end{cases}$$

Solve the whole-line problem with initial data $\widetilde{g}$ and $\widetilde{h}$. Since both extensions are odd, the solution remains odd in $x$, and hence

$$u(t, 0) = 0.$$

Using d'Alembert's formula,

$$u(t, x) = \frac{1}{2}\big[\widetilde{g}(x - t) + \widetilde{g}(x + t)\big] + \frac{1}{2}\int_{x-t}^{x+t} \widetilde{h}(y)\, dy.$$

For $x > t$, this becomes

$$u(t, x) = \frac{1}{2}\big[g(x - t) + g(x + t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy.$$

For $0 < x < t$,

$$u(t, x) = \frac{1}{2}\big[g(x + t) - g(t - x)\big] + \frac{1}{2}\int_{t-x}^{t+x} h(y)\, dy.$$

Thus the odd extension automatically produces the reflection required by the zero boundary condition.

## 5.6 Method of Spherical Means

We now consider

$$\begin{cases} u_{tt} - \Delta u = 0, & t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = g(x), \\ u_t(0,x) = h(x). \end{cases}$$

For fixed $x \in \mathbb{R}^d$, define the spherical integrals

$$U(t,r) = \int_{\partial B_r(x)} u(t,y)\, dS_y,$$

$$G(r) = \int_{\partial B_r(x)} g(y)\, dS_y,$$

and

$$H(r) = \int_{\partial B_r(x)} h(y)\, dS_y.$$

The key observation is that spherical averaging reduces the $d$-dimensional wave equation to an equation involving only $t$ and $r$.

**Lemma 5.6.1** (Euler–Poisson–Darboux Equation)**.** For fixed $x \in \mathbb{R}^d$, the spherical mean of a solution of the wave equation satisfies an equation of the form

$$U_{tt} - U_{rr} - \frac{d-1}{r} U_r = 0.$$

*Sketch of Proof.* Differentiate the spherical average with respect to $t$:

$$U_{tt}(t,r) = \int_{\partial B_r(x)} u_{tt}(t,y)\, dS_y.$$

Since $u_{tt} = \Delta u$,

$$U_{tt}(t,r) = \int_{\partial B_r(x)} \Delta u(t,y)\, dS_y.$$

The divergence theorem and differentiation with respect to the radius convert this expression into

$$U_{rr} + \frac{d-1}{r} U_r.$$

Hence

$$U_{tt} - U_{rr} - \frac{d-1}{r} U_r = 0.$$

□

For radial functions $\phi(x) = \phi(r)$, the same identity appears as

$$\Delta \phi = \phi''(r) + \frac{d-1}{r} \phi'(r).$$

## 5.7 Wave Equation in Three Dimensions

In dimension $d = 3$, the spherical mean method leads to Kirchhoff's formula.

**Theorem 5.7.1** (Kirchhoff Formula)**.** For the three-dimensional wave equation,

$$\begin{cases} u_{tt} - \Delta u = 0, \\ u(0,x) = g(x), \\ u_t(0,x) = h(x), \end{cases}$$

the solution is

$$u(t,x) = \frac{\partial}{\partial t}\left[\frac{1}{4\pi t}\int_{\partial B_t(x)} g(y)\, dS_y\right] + \frac{1}{4\pi t}\int_{\partial B_t(x)} h(y)\, dS_y.$$

Thus in three dimensions, the solution at $(t,x)$ depends only on the initial data on the sphere

$$\partial B_t(x).$$

**Remark 5.7.2** (Huygens' Principle)**.** In three spatial dimensions, the influence of an initial disturbance is concentrated on the advancing wave front. The solution at $(t,x)$ depends on initial data exactly at distance $t$ from $x$, not on the entire interior ball.

## 5.8 Wave Equation in Two Dimensions

In two dimensions the representation formula is different.

**Theorem 5.8.1** (Poisson Formula)**.** For the two-dimensional wave equation,

$$\begin{cases} u_{tt} - \Delta u = 0, \\ u(0,x) = g(x), \\ u_t(0,x) = h(x), \end{cases}$$

the solution is

$$u(t,x) = \frac{\partial}{\partial t}\left[\frac{1}{2\pi}\int_{B_t(x)} \frac{g(y)}{\sqrt{t^2 - |x-y|^2}}\, dy\right] + \frac{1}{2\pi}\int_{B_t(x)} \frac{h(y)}{\sqrt{t^2 - |x-y|^2}}\, dy.$$

Unlike the three-dimensional formula, the two-dimensional solution depends on the entire ball $B_t(x)$, not merely its boundary.
The exact representation therefore depends strongly on the spatial dimension, although finite propagation speed remains valid.

## 5.9 Nonhomogeneous Wave Equation

Consider

$$\begin{cases} u_{tt} - \Delta u = f(t,x), \quad t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = 0, \\ u_t(0,x) = 0. \end{cases}$$

As for the heat equation, a forcing term can be treated by Duhamel's principle.

For every $s \geq 0$, let $v(t, x; s)$ solve

$$\begin{cases} v_{tt} - \Delta v = 0, & t > s, \\ v(s, x; s) = 0, \\ v_t(s, x; s) = f(s, x). \end{cases}$$

Define

$$u(t, x) = \int_0^t v(t, x; s)\, ds.$$

**Theorem 5.9.1** (Duhamel's Principle for the Wave Equation)**.** The function above satisfies

$$u_{tt} - \Delta u = f$$

with zero initial data.

*Proof.* Differentiate:

$$u_t(t, x) = v(t, x; t) + \int_0^t v_t(t, x; s)\, ds.$$

Since $v(t, x; t) = 0$,

$$u_t(t, x) = \int_0^t v_t(t, x; s)\, ds.$$

Differentiating again,

$$u_{tt}(t, x) = v_t(t, x; t) + \int_0^t v_{tt}(t, x; s)\, ds.$$

By construction,

$$v_t(t, x; t) = f(t, x),$$

and since $v_{tt} = \Delta v$,

$$u_{tt} = f + \int_0^t \Delta v(t, x; s)\, ds.$$

Hence

$$u_{tt} = f + \Delta u,$$

so

$$u_{tt} - \Delta u = f.$$

□

In one dimension, d'Alembert's formula gives

$$v(t, x; s) = \frac{1}{2} \int_{x-(t-s)}^{x+(t-s)} f(s, y)\, dy.$$

Therefore

$$u(t, x) = \frac{1}{2} \int_0^t \int_{x-(t-s)}^{x+(t-s)} f(s, y)\, dy\, ds.$$

The integration region is the backward characteristic cone

$$0 \leq s \leq t, \qquad |x - y| \leq t - s.$$

This again shows finite propagation speed: only forcing inside the backward cone can affect the point $(t, x)$.

## 5.10 Energy Method and Uniqueness

We now study the wave equation without relying on an explicit solution formula. Consider

$$\begin{cases} u_{tt} - \Delta u = f, & (t,x) \in \Omega_T, \\ u = g, & (t,x) \in \partial\Omega \times (0,T), \\ u(0,x) = u_0(x), \\ u_t(0,x) = u_1(x). \end{cases}$$

To prove uniqueness, suppose $u_1$ and $u_2$ are two solutions with the same data and define

$$v = u_1 - u_2.$$

Then

$$v_{tt} - \Delta v = 0$$

with zero initial and boundary data.
Define the energy

$$E(t) = \frac{1}{2} \int_\Omega \left( |v_t|^2 + |\nabla v|^2 \right) \, dx.$$

Then

$$E'(t) = \int_\Omega \left( v_t v_{tt} + \nabla v \cdot \nabla v_t \right) \, dx.$$

Integration by parts gives

$$\int_\Omega \nabla v \cdot \nabla v_t \, dx = - \int_\Omega (\Delta v) v_t \, dx + \int_{\partial\Omega} v_t \frac{\partial v}{\partial \nu} \, dS.$$

The boundary term vanishes for homogeneous Dirichlet data. Therefore

$$E'(t) = \int_\Omega v_t (v_{tt} - \Delta v) \, dx = 0.$$

Thus

$$E(t) = E(0) = 0.$$

Hence

$$v_t = 0, \qquad \nabla v = 0,$$

and the zero initial or boundary data imply $v = 0$.

**Theorem 5.10.1** (Uniqueness)**.** The classical solution of the wave initial-boundary value problem, if it exists, is unique.

**Remark 5.10.2.** For the homogeneous wave equation, energy is conserved rather than dissipated:

$$E(t) = E(0).$$

This is another major difference from the heat equation, whose natural energy decreases with time.

## 5.11 Domain of Dependence

The energy method can also prove finite propagation speed without using an explicit formula.

Fix $t_0 > 0$ and $x_0 \in \mathbb{R}^d$. Define the backward cone

$$C = \{(t, x) : 0 \le t \le t_0,\ |x - x_0| \le t_0 - t\}.$$

**Proposition 5.11.1** (Domain of Dependence)**.** Suppose

$$u_{tt} - \Delta u = 0$$

in $C$, and

$$u(0, x) = u_t(0, x) = 0$$

for $x \in B_{t_0}(x_0)$. Then

$$u = 0$$

throughout $C$.

*Sketch of Proof.* Define the localized energy

$$e(t) = \frac{1}{2} \int_{B_{t_0 - t}(x_0)} \left( |u_t(t, x)|^2 + |\nabla u(t, x)|^2 \right) dx.$$

Because the domain of integration shrinks with time, differentiating $e(t)$ produces both an interior term and a boundary term. Using

$$u_{tt} - \Delta u = 0$$

and integration by parts, one obtains

$$e'(t) \le 0.$$

Since the initial data vanish in $B_{t_0}(x_0)$,

$$e(0) = 0.$$

Because $e(t) \ge 0$,

$$e(t) = 0$$

for $0 \le t \le t_0$. Therefore $u = 0$ throughout the cone. □

The proposition means that the value of the solution at $(t_0, x_0)$ depends only on data inside the backward cone.

## 5.12 Energy Estimate for a Forced Wave Equation

Suppose

$$u_{tt} - \Delta u = f.$$

For the localized energy above, the same calculation gives an inequality of the form

$$e'(t) \le e(t) + \frac{1}{2} \int_{B_{t_0 - t}(x_0)} |f(t, x)|^2 \, dx.$$

Applying Gronwall's inequality yields

$$e(t) \le e^t \left[ e(0) + \frac{1}{2} \int_0^t \int_{B_{t_0 - s}(x_0)} |f(s, x)|^2 \, dx \, ds \right].$$

Thus the energy inside a cone is controlled only by the initial energy and the forcing occurring inside the corresponding backward cone.

## 5.13 Separation of Variables

Consider the wave equation on $(0, \pi)$:

$$\begin{cases} u_{tt} - u_{xx} = 0, & t > 0,\ 0 < x < \pi, \\ u(t,0) = u(t,\pi) = 0, & t > 0, \\ u(0,x) = \phi(x), \\ u_t(0,x) = \psi(x). \end{cases}$$

Assume

$$u(t,x) = T(t)X(x).$$

Substitution gives

$$T''X - TX'' = 0.$$

Dividing by $TX$,

$$\frac{T''}{T} = \frac{X''}{X} = -\lambda.$$

Thus

$$T'' + \lambda T = 0$$

and

$$\begin{cases} -X'' = \lambda X, \\ X(0) = X(\pi) = 0. \end{cases}$$

The eigenvalues and eigenfunctions are

$$\lambda_n = n^2, \qquad X_n(x) = \sin(nx), \qquad n = 1, 2, \ldots.$$

Hence

$$u(t,x) = \sum_{n=1}^{\infty} T_n(t) \sin(nx),$$

where

$$T_n'' + n^2 T_n = 0.$$

Expand the initial data as Fourier sine series:

$$\phi(x) = \sum_{n=1}^{\infty} \phi_n \sin(nx),$$

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n \sin(nx),$$

with

$$\phi_n = \frac{2}{\pi} \int_0^{\pi} \phi(x) \sin(nx)\, dx,$$

and

$$\psi_n = \frac{2}{\pi} \int_0^{\pi} \psi(x) \sin(nx)\, dx.$$

The temporal coefficients satisfy

$$T_n(0) = \phi_n, \qquad T_n'(0) = \psi_n.$$

Therefore

$$T_n(t) = \phi_n \cos(nt) + \frac{\psi_n}{n} \sin(nt).$$

Hence

$$u(t,x) = \sum_{n=1}^{\infty} \left[ \phi_n \cos(nt) + \frac{\psi_n}{n} \sin(nt) \right] \sin(nx).$$

Unlike the heat equation, the coefficients do not decay exponentially. Instead, each spatial mode oscillates indefinitely.

## 5.14 Resonance

Consider the forced problem

$$\begin{cases} u_{tt} - u_{xx} = f(t,x), & t > 0,\ 0 < x < \pi, \\ u(t,0) = u(t,\pi) = 0, \\ u(0,x) = 0, \\ u_t(0,x) = 0. \end{cases}$$

Expand

$$f(t,x) = \sum_{n=1}^{\infty} f_n(t) \sin(nx).$$

Then

$$u(t,x) = \sum_{n=1}^{\infty} T_n(t) \sin(nx),$$

where

$$T_n'' + n^2 T_n = f_n(t).$$

Suppose the forcing is periodic in time and concentrated in one mode:

$$f(t,x) = A_n \sin(\omega t) \sin(nx).$$

Then

$$T_n'' + n^2 T_n = A_n \sin(\omega t).$$

By Duhamel's principle,

$$T_n(t) = \frac{A_n}{n} \int_0^t \sin(\omega s) \sin\Big(n(t-s)\Big)\, ds.$$

If $\omega \neq n$, this remains bounded and consists of oscillatory terms with frequencies $n$ and $\omega$.
If

$$\omega = n,$$

the forcing frequency agrees with the natural frequency of the $n$-th mode. In this case,

$$T_n(t) = -\frac{A_n}{2n^2} \sin(nt) + \frac{A_n}{2n} t \cos(nt),$$

up to an equivalent choice of phase convention.
The important term is

$$t \cos(nt),$$

whose amplitude grows linearly in time.

**Definition 5.14.1** (Resonance)**.** Resonance occurs when the frequency of an external periodic forcing matches a natural frequency of the system. The corresponding mode can then acquire an amplitude that grows with time.

For the interval $(0, \pi)$ with homogeneous Dirichlet boundary conditions, the natural frequencies are

$$1, 2, 3, \ldots.$$

Thus a forcing frequency $\omega = n$ resonates with the $n$-th spatial mode.

## 5.15 Summary

The wave equation exhibits several features that distinguish it from diffusion equations. In one dimension, d'Alembert's formula

$$u(t,x) = \frac{1}{2}\big[g(x-t) + g(x+t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy$$

shows directly that information travels along characteristic directions.
In higher dimensions, spherical means lead to dimension-dependent representation formulas. Duhamel's principle incorporates external forcing by superposing homogeneous waves produced at different times.
The conserved energy

$$E(t) = \frac{1}{2}\int_\Omega \left(|u_t|^2 + |\nabla u|^2\right)\,dx$$

provides uniqueness and gives another proof of finite propagation speed.
Finally, separation of variables decomposes a bounded-domain wave into independent oscillatory modes. When an external forcing matches one of their natural frequencies, resonance occurs.

# Chapter 6

# Hamilton–Jacobi Equations

Hamilton–Jacobi equations are nonlinear first-order partial differential equations of the form

$$u_t + H(\nabla u, x) = f(t, x),$$

or, in the simplest autonomous case,

$$u_t + H(\nabla u) = 0.$$

They are closely related to the method of characteristics, classical mechanics, the calculus of variations, and conservation laws. In contrast to linear transport equations, the characteristic velocity generally depends on the gradient of the unknown solution itself.

## 6.1 Hamilton–Jacobi Equations and Characteristics

Consider

$$u_t + H(\nabla u, x) = f(t, x).$$

For simplicity, first consider the homogeneous equation

$$u_t + H(\nabla u) = 0.$$

Set

$$p = \nabla u.$$

Differentiating the equation with respect to $x$,

$$p_t + D_p H(p) \cdot \nabla p = 0.$$

This has the form of a transport equation for $p$. Therefore the characteristic curves satisfy

$$\dot{x}(t) = D_p H(p(t)).$$

Along a characteristic,

$$\frac{d}{dt} p(t, x(t)) = p_t + \dot{x} \cdot \nabla p = 0.$$

Hence $p$ is constant along the characteristic.
Since

$$p = \nabla u,$$

we also compute

$$\frac{d}{dt} u(t, x(t)) = u_t + \dot{x} \cdot \nabla u.$$

Using

$$u_t = -H(p)$$

and

$$\dot{x} = D_p H(p),$$

we obtain

$$\frac{d}{dt} u(t, x(t)) = p \cdot D_p H(p) - H(p).$$

Thus the characteristic system is

$$\begin{cases} \dot{x} = D_p H(p), \\ \dot{p} = 0, \\ \dot{u} = p \cdot D_p H(p) - H(p). \end{cases}$$

More generally, if

$$H = H(p, x),$$

then the characteristic system becomes

$$\begin{cases} \dot{x} = D_p H(p, x), \\ \dot{p} = -D_x H(p, x), \\ \dot{u} = p \cdot D_p H(p, x) - H(p, x). \end{cases}$$

These are Hamilton's equations together with the evolution equation for $u$.

**Proposition 6.1.1** (Conservation of the Hamiltonian)**.** Suppose $H = H(p, x)$ has no explicit dependence on time. Along a Hamiltonian trajectory,

$$\dot{x} = D_p H, \qquad \dot{p} = -D_x H,$$

the Hamiltonian is constant:

$$H(p(t), x(t)) = H(p(0), x(0)).$$

*Proof.* By the chain rule,

$$\frac{d}{dt} H(p(t), x(t)) = D_p H \cdot \dot{p} + D_x H \cdot \dot{x}.$$

Substituting Hamilton's equations,

$$\frac{d}{dt} H = D_p H \cdot (-D_x H) + D_x H \cdot D_p H = 0.$$

Hence $H$ is constant along the trajectory. □

**Remark 6.1.2.** This is the Hamilton–Jacobi analogue of the characteristic method used for transport equations. The essential difference is that the characteristic velocity is now determined by the gradient $p = \nabla u$.

## 6.2 Calculus of Variations and Lagrangian Mechanics

Hamilton–Jacobi equations arise naturally from optimization problems.
Let

$$I[x] = \int_0^T L(x(t), \dot{x}(t))\, dt,$$

where $L = L(x, v)$ is called the *Lagrangian*.
Suppose $x(t)$ minimizes $I$ among curves with prescribed endpoints. Consider a variation

$$x_\varepsilon(t) = x(t) + \varepsilon \eta(t),$$

where

$$\eta(0) = \eta(T) = 0.$$

Since $x$ is a minimizer,

$$\frac{d}{d\varepsilon} I[x_\varepsilon]\Big|_{\varepsilon=0} = 0.$$

Differentiating,

$$0 = \int_0^T \left( D_x L \cdot \eta + D_v L \cdot \dot{\eta} \right) \, dt.$$

Integrating the second term by parts,

$$\int_0^T D_v L \cdot \dot{\eta} \, dt = - \int_0^T \frac{d}{dt} D_v L \cdot \eta \, dt,$$

since the boundary terms vanish.
Therefore

$$\int_0^T \left[ D_x L - \frac{d}{dt} D_v L \right] \cdot \eta \, dt = 0.$$

Since this holds for every admissible variation $\eta$,

$$\frac{d}{dt} D_v L = D_x L.$$

**Theorem 6.2.1** (Euler–Lagrange Equation)**.** If $x(t)$ is a smooth minimizer of

$$I[x] = \int_0^T L(x, \dot{x}) \, dt,$$

then

$$\frac{d}{dt} D_v L(x, \dot{x}) = D_x L(x, \dot{x}).$$

A standard example from mechanics is

$$L(x, v) = \frac{1}{2} m \, |v|^2 - V(x),$$

where the first term is kinetic energy and $V(x)$ is potential energy.
Then

$$D_v L = mv, \qquad D_x L = -\nabla V.$$

The Euler–Lagrange equation gives

$$m\ddot{x} = -\nabla V(x),$$

which is Newton's second law.
Thus the calculus of variations provides a bridge between mechanics and Hamilton–Jacobi equations.

## 6.3 Legendre Transform

The Hamiltonian is obtained from the Lagrangian by a Legendre transform.

**Definition 6.3.1** (Legendre Transform)**.** Let $L : \mathbb{R}^d \to \mathbb{R}$ be convex. Its Legendre transform is

$$L^*(p) = \sup_{v \in \mathbb{R}^d} \Big\{ p \cdot v - L(v) \Big\}.$$

In classical mechanics, we define the momentum

$$p = D_v L(v)$$

and the Hamiltonian

$$H(p) = p \cdot v - L(v),$$

where $v$ is expressed in terms of $p$.
Thus

$$H = L^*.$$

Conversely, if $H$ is convex, then

$$L = H^*.$$

**Example 6.3.2.** Let

$$L(v) = \frac{1}{2} |v|^2 .$$

Then

$$L^*(p) = \sup_v \left\{ p \cdot v - \frac{1}{2} |v|^2 \right\}.$$

The maximum occurs at $v = p$, and therefore

$$L^*(p) = \frac{1}{2} |p|^2 .$$

Thus

$$H(p) = \frac{1}{2} |p|^2 .$$

The basic inequality behind the Legendre transform is the following.

**Lemma 6.3.3** (Young–Fenchel Inequality)**.** For every $p, v \in \mathbb{R}^d$,

$$p \cdot v \leq H(p) + L(v).$$

Indeed, since

$$H(p) = \sup_v \{p \cdot v - L(v)\},$$

we have

$$H(p) \geq p \cdot v - L(v).$$

Rearranging gives the result.
When equality holds,

$$p = D_v L(v),$$

or equivalently,

$$v = D_p H(p),$$

under suitable differentiability and strict convexity assumptions.

**Remark 6.3.4.** The Legendre transform converts the velocity variable of the Lagrangian description into the momentum variable of the Hamiltonian description. This is why the same transform appears naturally both in mechanics and in Hamilton–Jacobi equations.

## 6.4 Hopf–Lax Formula

Consider the Hamilton–Jacobi initial-value problem

$$\begin{cases} u_t + H(\nabla u) = 0, & t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = g(x), & x \in \mathbb{R}^d, \end{cases}$$

where $H$ is convex.
Let $L = H^*$ be the Legendre transform of $H$.
The solution can be represented by an optimization formula.

**Theorem 6.4.1** (Hopf–Lax Formula)**.** Under suitable assumptions on $H$ and $g$,

$$u(t,x) = \inf_{y \in \mathbb{R}^d} \left\{ tL\left(\frac{x-y}{t}\right) + g(y) \right\}.$$

The formula has a direct variational interpretation. To arrive at $(t,x)$, imagine starting from a point $y$ at time 0. The initial cost is $g(y)$, while

$$tL\left(\frac{x-y}{t}\right)$$

is the cost of travelling from $y$ to $x$ with constant velocity

$$\frac{x-y}{t}.$$

The solution chooses the least expensive starting point.
Equivalently,

$$u(t,x) = \inf_{y \in \mathbb{R}^d} \left\{ g(y) + tL\left(\frac{x-y}{t}\right) \right\}.$$

**Example 6.4.2.** For

$$H(p) = \frac{1}{2}|p|^2,$$

we also have

$$L(v) = \frac{1}{2}|v|^2.$$

Therefore

$$u(t,x) = \inf_{y \in \mathbb{R}^d} \left\{ g(y) + \frac{|x-y|^2}{2t} \right\}.$$

We now sketch why the Hopf–Lax formula is consistent with the Hamilton–Jacobi equation.
Suppose the infimum is attained at $y = y(t,x)$. Then

$$u(t,x) = tL\left(\frac{x-y}{t}\right) + g(y).$$

Set

$$v = \frac{x-y}{t}.$$

Differentiating formally with respect to $x$,

$$\nabla u = D_v L(v).$$

Hence, by Legendre duality,

$$v = D_p H(\nabla u).$$

Differentiating with respect to $t$,

$$u_t = L(v) - D_v L(v) \cdot v.$$

Since

$$H(\nabla u) = \nabla u \cdot v - L(v),$$

we obtain

$$u_t + H(\nabla u) = 0.$$

**Remark 6.4.3.** The method of characteristics describes the solution locally by following trajectories, while the Hopf–Lax formula describes it globally through a minimization problem. Both are manifestations of the same Hamiltonian structure.

## 6.5 Dynamic Programming and General Hamiltonians

The Hopf–Lax formula satisfies a semigroup or dynamic programming property.
Let

$$S_t g(x) = \inf_{y \in \mathbb{R}^d} \left\{ tL\left(\frac{x-y}{t}\right) + g(y) \right\}.$$

Then

$$S_{t+s} g = S_t(S_s g).$$

This means that an optimal path from time 0 to time $t+s$ may be split at an intermediate time $s$, and the optimization may be performed in two stages.
For a more general equation

$$u_t + H(\nabla u, x) = f(t, x),$$

the variational representation takes the form

$$u(t,x) = \inf_{\gamma(t)=x} \left\{ g(\gamma(0)) + \int_0^t \left[ L(\dot{\gamma}(s), \gamma(s)) + f(s, \gamma(s)) \right] ds \right\},$$

with signs adjusted according to the convention used for the forcing term.
Here the infimum is taken over admissible curves $\gamma : [0, t] \to \mathbb{R}^d$.
Thus the solution represents an optimal accumulated cost:

$$\text{initial cost} + \text{running cost}.$$

**Remark 6.5.1.** This variational interpretation is one reason Hamilton–Jacobi equations appear naturally in optimal control and classical mechanics.

## 6.6 Basic Estimates for the Hopf–Lax Formula

Suppose $g$ is Lipschitz:

$$|g(x) - g(y)| \leq C\,|x - y|\,.$$

Under suitable growth assumptions on $L$, the Hopf–Lax formula preserves spatial Lipschitz regularity.

To see the basic idea, write

$$u(t,x) = \inf_y \left\{ tL\left(\frac{x-y}{t}\right) + g(y) \right\}.$$

For $x_1, x_2 \in \mathbb{R}^d$, compare the two minimization problems using translated competitors. This gives an estimate of the form

$$|u(t,x_1) - u(t,x_2)| \leq C\,|x_1 - x_2|\,.$$

Under additional assumptions, one can also obtain Lipschitz estimates in time:

$$|u(t,x) - u(s,x)| \leq C\,|t-s|\,.$$

Thus even when a classical differentiable solution eventually fails to exist, the variational formula can still produce a well-behaved continuous solution.

## 6.7 Semiconcavity

Another useful regularity property is semiconcavity.

**Definition 6.7.1** (Semiconcavity)**.** A function $f : \mathbb{R}^d \to \mathbb{R}$ is called semiconcave if there exists $C \geq 0$ such that

$$f(x+z) + f(x-z) - 2f(x) \leq C\,|z|^2$$

for all relevant $x, z$.

Equivalently, when $f$ is twice differentiable, semiconcavity corresponds formally to an upper bound on the Hessian:

$$D^2 f \leq CI.$$

For the Hopf–Lax formula, assumptions on the convexity of $H$ or $L$ often imply that $u(t,\cdot)$ becomes semiconcave for $t > 0$.
Semiconcavity is weaker than smoothness but strong enough to control the behavior of first derivatives and plays an important role in first-order nonlinear PDEs.

## 6.8 Conservation Laws

Hamilton–Jacobi equations are closely related to scalar conservation laws.
Consider in one dimension

$$u_t + H(u_x) = 0.$$

Set

$$v = u_x.$$

Differentiating with respect to $x$,

$$v_t + \partial_x H(v) = 0.$$

Thus $v$ satisfies the conservation law

$$v_t + \Big(H(v)\Big)_x = 0.$$

More generally, a scalar conservation law has the form

$$u_t + \nabla \cdot F(u) = 0,$$

where $F(u)$ is called the flux.
The word "conservation" comes from integrating over a region. In one dimension,

$$u_t + F(u)_x = 0.$$

Integrating over an interval $[a, b]$,

$$\frac{d}{dt}\int_a^b u(t,x)\,dx = F(u(t,a)) - F(u(t,b)).$$

Hence the total amount inside the interval changes only through the flux across its boundary.
This is the same conservation principle used in the Introduction, but the flux is now allowed to depend nonlinearly on the unknown $u$.

## 6.9 Weak Solutions of Conservation Laws

Classical solutions of nonlinear conservation laws may develop discontinuities even from smooth initial data. We therefore use an integrated formulation.
Consider

$$u_t + \nabla \cdot F(u) = 0.$$

Multiply by a test function

$$\varphi \in C_c^\infty((0,\infty)\times\mathbb{R}^d)$$

and integrate:

$$\int_0^\infty \int_{\mathbb{R}^d} [u_t\varphi + \nabla \cdot F(u)\,\varphi]\,dx\,dt = 0.$$

Integration by parts gives

$$\int_0^\infty \int_{\mathbb{R}^d} [u\varphi_t + F(u)\cdot\nabla\varphi]\,dx\,dt = 0.$$

**Definition 6.9.1** (Weak Solution of a Conservation Law)**.** A locally integrable function $u$ is a weak solution of

$$u_t + \nabla \cdot F(u) = 0$$

if

$$\int_0^\infty \int_{\mathbb{R}^d} [u\varphi_t + F(u)\cdot\nabla\varphi]\,dx\,dt = 0$$

for every $\varphi \in C_c^\infty((0,\infty)\times\mathbb{R}^d)$.

The advantage is that the definition remains meaningful even when $u$ has jump discontinuities.

## 6.10 Rankine–Hugoniot Condition

Suppose a one-dimensional weak solution has a jump discontinuity across a curve

$$x = s(t).$$

Let

$$u_L$$

and

$$u_R$$

denote the values to the left and right of the discontinuity.
The speed of the discontinuity cannot be arbitrary. Conservation requires a relation between the jump in $u$ and the jump in flux.

**Theorem 6.10.1** (Rankine–Hugoniot Condition)**.** For the conservation law

$$u_t + F(u)_x = 0,$$

a discontinuity moving with speed $s'(t)$ must satisfy

$$s'(t)\Big(u_R - u_L\Big) = F(u_R) - F(u_L).$$

If $u_L \neq u_R$, then

$$s'(t) = \frac{F(u_R) - F(u_L)}{u_R - u_L}.$$

*Sketch of Proof.* Consider a thin space-time region surrounding the discontinuity curve. Integrating

$$u_t + F(u)_x = 0$$

over this region and applying the divergence theorem gives conservation of the total flux across its boundary.
Letting the width of the region tend to zero leaves only the contributions from the two sides of the moving discontinuity. The resulting balance is

$$s'(t)(u_R - u_L) = F(u_R) - F(u_L).$$

□

It is convenient to use the jump notation

$$[u] = u_R - u_L, \qquad [F(u)] = F(u_R) - F(u_L).$$

Then the Rankine–Hugoniot condition becomes

$$s'[u] = [F(u)].$$

**Example 6.10.2** (Burgers' Equation)**.** Consider

$$u_t + uu_x = 0.$$

Since

$$uu_x = \left(\frac{u^2}{2}\right)_x,$$

the equation can be written in conservation form as

$$u_t + \left(\frac{u^2}{2}\right)_x = 0.$$

Thus

$$F(u) = \frac{u^2}{2}.$$

If a discontinuity connects constant states $u_L$ and $u_R$, then its speed is

$$s' = \frac{\frac{1}{2}u_R^2 - \frac{1}{2}u_L^2}{u_R - u_L} = \frac{u_L + u_R}{2}.$$

This also explains the characteristic crossing encountered earlier for Burgers' equation. Once characteristics intersect, the classical solution breaks down, but a discontinuous weak solution may continue beyond the breakdown time.

**Remark 6.10.3.** The Rankine–Hugoniot condition is a necessary conservation condition for a discontinuity. It does not by itself determine which weak solution is physically relevant when several weak solutions exist. The additional selection theory is beyond the scope of these notes.

## 6.11 Summary

Hamilton–Jacobi equations connect several themes developed throughout these notes. The method of characteristics transforms

$$u_t + H(\nabla u, x) = 0$$

into Hamilton's system

$$\dot{x} = D_p H, \qquad \dot{p} = -D_x H.$$

The same structure arises from the calculus of variations through the Euler–Lagrange equation.

The Legendre transform connects the Lagrangian and Hamiltonian descriptions:

$$H = L^*, \qquad L = H^*.$$

For convex Hamiltonians, the Hopf–Lax formula represents the solution as an optimization problem:

$$u(t,x) = \inf_y \left\{ g(y) + tL\left(\frac{x-y}{t}\right) \right\}.$$

Finally, differentiating a Hamilton–Jacobi equation in space naturally leads to a conservation law. When discontinuities form, the classical PDE must be replaced by a weak formulation, and the motion of a jump is governed by the Rankine–Hugoniot condition

$$s'[u] = [F(u)].$$

Thus characteristics, optimization, mechanics, and conservation laws are different expressions of the same underlying first-order nonlinear structure.

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These notes were compiled from my handwritten lecture notes and subsequent review. Their purpose is not to serve as a comprehensive textbook, but to preserve the main ideas, derivations, proofs, and methods of the course in a form that I can return to in the future.

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